Defensive Efficiency in Path of Exile 2: Physical Damage Mitigation Across the Ascendancy Spectrum

A Tier Ranking of Ascendancy Defensive Efficiency: Virtual HP, Optimal Armour, and the Opportunity Cost of Survivability

Path of Exile 2 — Defensive Layer Theory

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Contents

Executive Summary

- **The Armour Formula Has a Structural Ceiling.** Path of Exile 2's damage reduction formula (DR $= A/(A + 10H)$) degrades nonlinearly with hit size. At the endgame hit ceiling (H = 10,000), 100,000 armour yields only 50% DR — requiring 5,001 effective HP to survive. Doubling the armour budget to 200,000 raises DR to only 66.7%. The formula's denominator grows with H faster than any achievable armour investment can compensate. Every competitive archetype examined in this paper circumvents this ceiling rather than overcoming it.
- **Virtual HP: The Tier-Ranking Metric.** This paper ranks all archetypes by their Virtual HP multiplier, defined as VHP = eHP/(1 − DR%). VHP collapses the full defensive configuration — armour value, conversion, absorption, avoidance probability — to a single figure: how many effective hit points the archetype creates relative to the pool it invests in. A 2.0× VHP multiplier means the build survives hits twice as large as its raw HP would imply; a 5.2× multiplier means it survives hits 5.2× its pool. The baseline armour formula at optimal investment yields 2.0×. Every mechanism in this paper targets a higher multiple.
- **Binding Constraint: Conversion Releases the Armour Budget.** The character-sheet analysis in Section 3 establishes the paper's central diagnostic. A 0-armour Shaman — with 75% conversion and 85% elemental resistance caps but no armour investment whatsoever — takes 3,625 damage at H = 10,000, flat at every hit size. Smith of Kitava with 100,000 armour takes 5,000. Before Shaman has committed a single passive point to armour scaling, conversion alone outperforms the maximum-armour archetype at the hit sizes that actually kill characters. With optional 25,000 armour and armour-to-elemental from the ascendancy, the same Shaman takes 1,923 (80.8% total mitigation). Matching that figure through armour alone requires approximately 421,000 armour — beyond any achievable gear. The gap is structural: conversion suppresses the effective hit size before it enters the formula, making the same armour 4× more efficient per point. The 8–15 passive points Smith commits to armour scaling are freed on Shaman for damage, sustain, or additional defensive layers. This is not a performance margin — it is a structural release of the build's binding constraint.
- **Witchhunter: Sorcery Ward and the Absorption-Layer Ceiling.** The

Witchhunter (Mercenary ascendancy) achieves the highest single-hit survival ceiling of the archetypes examined in this paper. Two ascendancy notables (Obsessive Rituals + Ceremonial Ablution) convert armour and evasion investment into a Sorcery Ward absorption pool (SW = 30% of post-penalty armour+evasion, capped at 32,000) that intercepts all post-armour hit damage before it reaches life. At 60,000 geared armour (30,000 post-penalty) and L = 3,000 HP, SW = 9,000 — absorbing the full post-armour remainder of a 10,000 raw hit with zero life damage. The derived survival ceiling is approximately H = 14,485, substantially exceeding both Shaman and Smith at comparable passive investment. No unique item is required.

- **Infernalist: Chaos Inoculation and the Hard Immunity Advantage.** The Infernalist (Witch ascendancy) achieves comparable or superior mitigation through a distinct conversion stack: 50–55% physical-to-fire, 20% physical-to-chaos eliminated entirely via Chaos Inoculation, and a reduced physical residual of 20–30%. The CI keystone makes the chaos component a categorical zero rather than a near-zero residual, and simultaneously frees all chaos resistance affixes and life affixes on gear for Energy Shield and evasion. The Infernalist Full configuration (80% total conversion + CI) achieves 85.3% mitigation at H=10,000 — surpassing standard Shaman at 80.8% — while providing hard chaos immunity at every hit size.
- **Shaman: Conversion Stack (Corrected).** The Shaman converts 75% of incoming physical damage to elemental — 60% to fire (Cloak of Flame 50% + ascendancy 5% + passive tree notable 5%), 5% each to cold, lightning, and a random element. Only 25% remains physical. Prior analyses cited 60–70%; the additional passive tree notable raises the total to 75%, reducing residual physical by 37.5% and pushing total mitigation from the 63–70% range to 80.8% at the endgame hit ceiling.
- **Invoker: Avoidance-Primary Architecture and Conditional eHP Ceiling.** The Invoker (Monk ascendancy) achieves defensive efficiency through a fundamentally different paradigm: preventing hits from connecting via evasion and block, rather than suppressing damage on hits that land. The "...and protect me from harm" ascendancy node unifies the evasion and armour pools, applying 65% of base evasion as effective armour, so that investment in evasion simultaneously improves avoidance chance and provides armour DR. At 50%+50% evasion and block, 75% of all hits are avoided before any mitigation layer applies. Against the hits that do land, Meditate — a pre-combat skill

that overflows Energy Shield to 1.5× — provides the eHP buffer. The result is an archetype with lower per-hit DR than Shaman (27–41% at H=10,000) but comparable or lower total encounter damage intake under optimal play.

- **Warbringer: Block-as-Mitigation and the Mapping Specialist.** The Warbringer (Marauder ascendancy) converts block into a front-loaded damage multiplier via Turtle Charm: blocked hits deal 20% of their original value to the player (80% less), raising the effective block cap to 75%. Wooden Wall adds a second front-loaded layer — 20% of damage taken goes to an active totem instead. Combined on a blocked hit: $0.20 \times 0.80 = 0.16H$ enters the armour formula. At A=100k, the expected mitigation across the probability chain (75% block) is 89.5% — the highest of any armour-anchored archetype in this paper. The binding constraint is the 25% of unblocked hits, which arrive at 80% of their original value and impose a 3,557 HP minimum survival requirement at H=10,000. The build excels in mapping content where the block distribution averages out; it is exposed to single-hit variance in boss encounters, partially offset by Turtle Charm's unique ability to block otherwise unblockable hits.
- **Tactician: Deflection, Capped DR, and High Variance.** The Tactician (Mercenary ascendancy) achieves armour efficiency through "Stay Light, Use Cover," which defends with 200% of armour — halving the armour required to reach any given DR — while hard-capping both evasion and damage reduction at 50%. "Polish that Gear" converts 20% of armour into deflection rating; deflection reduces a hit by 40% before armour evaluates. A deflected hit at the 50% DR cap takes 30% of its original value (70% effective DR), comparable to Shaman. An undeflected hit at the cap takes 50%, requiring 5,001 effective HP to survive a 10,000 raw hit — identical to the Titan baseline. Expected mitigation across the probability chain (25% effective evasion, 75% deflection with Glancing Blows) is 73.8% at H=10,000, trailing Shaman's deterministic 80.8%. The build is efficient when deflection fires; it is exposed when it does not.
- **Smith of Kitava: B-Tier — Armour, Conversion, and the Critical Strike Frontier.** The Smith of Kitava (Marauder ascendancy) extends armour's reach to the elemental components of converted hits through "X% of Armour applies to Elemental Damage" stacking (Y factor, stackable beyond 1.0) paired with 35% physical-to-elemental conversion. The Blade Catcher ascendancy node converts incoming critical strikes to base damage and defends those hits at 200% of armour rating — doubling effective armour against the hits that would otherwise be lethal. At standard endgame investment ($Y = 1.0$, $A = 80{,}000$),

Smith achieves 60.2% total mitigation and a 2.5× VHP multiplier, trailing S-Tier conversion archetypes in general content but dominating the critical-strike map frontier that no other archetype addressed in this paper matches.

- **Titan: C-Tier — The Armour Formula's Natural Floor.** The Titan (Warrior ascendancy) is the most direct expression of the "armour big, life big" paradigm and provides the paper's baseline reference. At 100,000 armour and 5,001 effective HP it survives the endgame ceiling hit by exactly one hit point, producing a 2.0× VHP multiplier. There is no suppression of effective hit size, no absorption pool, no avoidance layer. The Titan is a robust platform through mid-game and early endgame content where hit sizes remain in the formula's efficient range; it is not a competitive ceiling for the hardest content, and every archetype in this paper exceeds its 2.0× VHP floor.

1 Armour Mechanics: The DR Formula and Its Limits

Path of Exile 2's damage reduction framework rests on a single formula that governs every armour-based archetype in this paper. Understanding why it fails at scale is a prerequisite for understanding why every competitive defensive archetype employs a mechanism to work around that failure. This section establishes the formula, its structural weaknesses, and the armour optimisation corridor that defines the planning constraint for all builds examined here.

The Path of Exile 2 Armour Formula

$$\mathrm{DR}_{\mathrm{phys}} = \frac{A}{A + 10H}$$

where A is armour rating and H is the size of the incoming physical hit. The formula exhibits strongly diminishing returns: the denominator grows linearly with hit size, causing DR% to decay increasingly fast as H increases. This is armour's fatal structural weakness against large hits. The only way to counteract it without increasing A is to reduce H — which is precisely what conversion does.

When armour applies to elemental damage ("X% of Armour applies to Elemental Damage"), the same formula evaluates each elemental hit component independently:

$$\mathrm{DR}_{\mathrm{ele}} = \frac{A}{A + 10 \cdot H_{\mathrm{ele}}}$$

At 100% armour-to-elemental (achievable by Shaman through ascendancy nodes), the full armour value applies. Critically, the cold, lightning, and random elemental components are each only 5% of the raw hit — for a 10,000 raw hit, each is only 500 damage. At 25,000 armour:

$$\text{DR vs 500 ele hit} = \frac{25{,}000}{25{,}000 + 5{,}000} = 83.3\%$$

These small components are nearly eliminated by armour alone, before resistance even applies. Only the 60% fire component (6,000 at H=10,000) represents a meaningful elemental contribution.

What "Quadruples the Armour Pool" Means Precisely

The conversion reduces the effective physical hit to 25% of the raw value. Because the armour formula evaluates against hit size rather than raw damage, this is equivalent to multiplying armour rating by 4× for physical reduction purposes:

$$\frac{A}{A + 10 \times \frac{H}{4}} = \frac{4A}{4A + 10H} \quad \Longleftrightarrow \quad \text{effective armour} = 4 \times A$$

Explicitly at H=10,000:

$$\text{Shaman: } \frac{25{,}000}{25{,}000 + 2{,}500} \times \frac{1}{0.25} \approx \frac{100{,}000}{200{,}000} = 50\% \quad = \quad \text{Smith: } \frac{100{,}000}{200{,}000}$$

Both achieve 50% physical DR on their respective physical components. Shaman uses 25,000 armour; Smith uses 100,000 armour. The conversion did the work of 75,000 additional armour rating.

The Armour Equivalence for Total Mitigation

Matching the Shaman's *total* effective mitigation (80.8%) through armour alone, with no conversion and 75% resistance on any elemental incidental damage:

$$\frac{A}{A + 100{,}000} = 0.808 \quad \Rightarrow \quad A = \frac{0.808 \times 100{,}000}{1 - 0.808} \approx 421{,}000$$

421,000 armour is not achievable in Path of Exile 2. The mitigation ceiling of Shaman's 25,000 armour plus 75% conversion cannot be replicated through armour investment at any practical scale.

2 Virtual HP: The Unifying Metric

The One-Shot Ceiling

The armour formula establishes *damage reduction* — what fraction of a hit a character absorbs. But damage reduction alone does not determine survival. A character with 95% DR still dies if the 5% remainder exceeds the effective HP pool (eHP). The correct quantity to maximise is the largest single hit the character can survive. This quantity is called **Virtual HP**:

$$\text{Virtual HP} = \frac{\text{eHP}}{1 - \text{DR\%}}$$

A character survives any single hit $H \leq$ Virtual HP. Virtual HP is the one-shot ceiling — the maximum incoming hit the character can absorb without dying.

The formula makes the binding constraint explicit: **eHP is what you are actually scaling**. DR% acts as a multiplier on eHP, converting the raw life pool into a larger effective survival ceiling, but a character with zero eHP cannot survive regardless of their DR%. Raising DR% without raising eHP produces a larger ceiling on a pool that may still be insufficient. Raising eHP while holding DR% constant scales the ceiling linearly. The two compound, but the formula's message is unambiguous: *life first*.

The Armour Formula's Native VHP Multiplier

Against a pure physical hit, the armour formula gives:

$$\text{DR} = \frac{A}{A + 10H}$$

At the planning target of $A = 20L$ evaluated against the kill threshold $H = 2L$ (where L is eHP and $H = 2L$ is the largest hit the build is designed to survive):

$$\text{DR} = \frac{20L}{20L + 20L} = 0.50 \quad \Rightarrow \quad \text{VHP} = \frac{L}{1 - 0.50} = 2L$$

A pure armour build at the $20 \times L$ corridor ceiling converts 1 unit of eHP into 2 units of Virtual HP — a 2.0× multiplier. This is the structural floor of what armour alone achieves. Every archetype examined in this paper is an attempt to exceed this floor through a mechanism the formula cannot naturally provide.

Every Archetype as a VHP Inflation Vehicle

The thesis of this paper is that all effective defensive archetypes in Path of Exile 2 share a common structural property: they inflate the VHP/eHP ratio beyond the armour baseline. The mechanisms differ — conversion suppresses the effective hit size, absorption intercepts post-DR damage, avoidance prevents hits from connecting — but the output metric is the same.

Table 1: **Virtual HP Multiplier by Archetype (Representative Values at H=10,000)**

Archetype	**Primary Mechanism**	**DR% at H=10k**	**VH**
Witchhunter	SW absorption (post-DR)	23.1% + SW buffer	Unbounded below H
Infernalist Full	Conversion + CI	85.3%	
Shaman	Physical-to-elemental conversion	80.8%	
Invoker	Avoidance + evasion-as-armour	27–41% per-hit	>5× encoun
Warbringer	Block + front-loaded multiplier	84.8% expected	6.6× exp /
Tactician	Deflection + capped DR	73.8% expected	3.8× e
Smith of Kitava	Armour-to-ele + crit frontier	60.2%	
Armour Pure / Titan	Raw armour and life only	50.0%	**2.0× (baselin**

VHP/eHP is the ratio by which each archetype inflates its survival ceiling above its raw eHP pool. Witchhunter's Sorcery Ward absorbs post-DR damage before life is reached, making VHP effectively unbounded below the SW ceiling. Warbringer and Tactician carry two entries because their VHP is probabilistic: the expected outcome is strong but the worst-case falls to the armour-pure floor on unfavourable rolls. The bottom row (Armour Pure / Titan at 2.0×) is the structural baseline all other archetypes are measured against.

The Opportunity Cost Dimension

VHP/eHP captures survival efficiency at a single investment snapshot. The paper's deeper argument is investment efficiency: how many passive points, gear affixes, and unique-item slots does each archetype spend to achieve its VHP multiplier?

- **Shaman** achieves 5.2× from a unique chest item (Cloak of Flame) and ascendancy notables. No passive tree armour investment is required.
- **Smith of Kitava** achieves 2.5× at the cost of heavy passive armour-node investment, dedicated armour-to-elemental node pathing, and Veil of Night as the unique enabler.
- **Titan** achieves 2.0× at the cost of the heaviest armour and life investment of any archetype, with no mechanism to exceed the structural floor.

The passive points Shaman does not spend on armour go to damage, sustain, or additional defensive layers. This is the opportunity cost argument: the comparison is not just VHP at a fixed investment level, but what each archetype foregoes to reach its VHP multiplier. The sections that follow rank archetypes from highest to lowest VHP ceiling, with explicit analysis of the passive and gear budget each invests to get there.

3 Character Sheet Deterministic Dynamics: Optimality, Opportunity Cost, and Efficient Attainment of Survival

The armour formula's failure at scale is most clearly demonstrated by comparing two archetypes often discussed as alternatives: the Smith of Kitava (dedicated armour scaling) and the Shaman (physical-to-elemental conversion). This section presents that comparison as a worked example of three claims central to the paper's thesis:

1. **Armour has a hard ceiling and a structural weakness** — PoE 2's physical DR cap of 90% bounds armour's advantage from above; the formula's quadratic degradation with hit size means DR collapses precisely when survival matters most.
2. **eHP is the binding constraint** — the minimum life pool required to survive a given hit is dictated by what the formula allows through; reducing that requirement via conversion is categorically more efficient than raising armour to compensate.
3. **Opportunity cost governs investment** — the passive points and gear affixes required to bridge the armour gap between Smith and Shaman cannot be recovered from other areas of the character sheet.

Build Parameters

Three configurations are compared throughout this section. **Shaman (0 armour)** is the true baseline: 75% physical-to-elemental conversion and 85% elemental resistance caps, with zero passive investment in armour nodes. This is the Shaman before any armour-specific decision has been made. **Smith of Kitava** is parameterised by the 12–20$\times L$ corridor: to survive $H = 10{,}000$, Smith requires at minimum $L = 5{,}001$ effective HP; at the optimal corridor ceiling ($A = 20 \times L_{\min} = 100{,}000$), Smith takes exactly 5,000 damage and survives with 1 HP. This is the *minimum viable* Smith configuration — any lower armour or life combination within the corridor fails. The **Baseline** (25,000 armour, no conversion) provides the armour formula's reference floor.

Table 2: **Build Parameters for Cross-Archetype Comparison**

Parameter	**Baseline**	**Shaman (0 arm)**	**Shaman (25k arm†)**	**Smith (corri**
Armour Rating	25,000	0	25,000	
Conversion (phys → ele)	0%	75%	75%	
Armour-to-Elemental	0%	0%	100% (asc.)	
Max Elemental Resistance	75%	85%	85%	
Min HP to survive H=10k	>8,001	3,625	1,923	
Armour investment	**1×**	**0×**	**1× (optional)**	**4× (ma**

†*Shaman (25k arm) is an optional investment: 25,000 armour plus 100% armour-to-elemental from Shaman ascendancy notables. Smith's 100,000 armour is derived from the corridor (*$A = 20 \times 5{,}000$*), not assumed; it is the minimum that keeps Smith alive at* $H = 10{,}000$*. PoE 2 caps physical DR at 90%; any formula output above 90% is delivered as 90%, not the formula value — this materially changes the small-hit comparison.*

The 0-Armour Shaman Baseline

With zero armour, the Shaman's defensive stack consists entirely of conversion and elemental resistance. Against any incoming physical hit H:

- Physical residual (25%): $0.25H$ — no damage reduction, taken at face value.
- Fire component (60%): $0.60H \times (1 - 0.85) = 0.09H$ — handled entirely by 85% resistance.
- Cold / lightning / random (5% each): $3 \times 0.05H \times 0.15 = 0.0225H$ — handled by 85% resistance.

Total damage taken $= (0.25 + 0.09 + 0.0225) \times H = 0.3625 \times H$. **Total mitigation: 63.75% at every hit size.** Because neither conversion percentages nor resistance caps change with H, the 0-armour Shaman's mitigation is a horizontal line — it does not degrade as hits scale.

At $H = 10{,}000$: $0.3625 \times 10{,}000 = 3{,}625$ damage taken. A Shaman without a single passive point in armour nodes requires 3,625 effective HP to survive the endgame ceiling hit. Smith with 100,000 armour requires 5,001. *The 0-armour conversion archetype already out-survives the maximum-armour archetype at the hit size that actually kills characters* — before any armour has been added to the Shaman build.

Table 3: **Total Effective Mitigation (%) by Incoming Hit Size — 90% DR Cap Applied**

Incoming Hit	Baseline (25k)	Smith (100k)	Shaman (0 arm)	Shaman (25k arm†)	Smith
100	90.0%*	90.0%*	63.8%	96.4%	
500	83.3%	90.0%*	63.8%	96.3%	
1,000	71.4%	90.0%*	63.8%	95.5%	
2,000	55.6%	83.3%	63.8%	92.7%	
4,000	38.5%	71.4%	63.8%	88.2%	
5,686	28.5%	63.8%	63.8%	85.8%	
8,000	23.8%	55.6%	63.8%	82.7%	S
10,000	20.0%	50.0%	63.8%	80.8%	Sh

**DR cap applied: PoE 2 caps physical DR at 90%. Smith's formula returns 99.0% at H=100, 95.2% at H=500, and 90.9% at H=1,000 — all floored to 90%; the formula values are not delivered. Baseline is similarly capped below H≈278. Shaman (0 arm) mitigation is flat at 63.75% at all hit sizes: no armour means no hit-size-dependent term. Shaman (25k arm†) applies the 90% cap per component; the composite total exceeds 90% because 85% resistance stacks multiplicatively on top of the capped armour DR for each elemental type. Cell shading: dark green ≥95%, light green 90–94%, yellow 80–89%, orange 70–79%, red <70%.*

Effective Mitigation by Hit Size

Absolute Damage Taken by Hit Size

Armour Leads at Low Hits: The 90% DR Cap Window

Below $H \approx 1{,}111$, Smith's armour formula output exceeds the 90% physical DR ceiling and is delivered as 90%. Both Smith and the Baseline are simultaneously at the cap:

$$H = 100: \quad \frac{100{,}000}{100{,}000 + 1{,}000} = 99.0\% \xrightarrow{\text{cap}} 90.0\% \qquad \frac{25{,}000}{25{,}000 + 250} = 99.0\% \xrightarrow{\text{cap}} 90.0\%$$

In the capped regime, Smith's 4× armour advantage over the Baseline produces zero additional benefit — both land at 90% DR regardless. The formula *feels* infallible at small hits, and within the cap it is. The cap also means Smith's physical takes are: 10 damage at H=100, 50 at H=500, 100 at H=1,000 — not the 1, 24, 91 values the uncapped formula would imply.

The 0-armour Shaman is at 63.75% in this regime — meaningfully below Smith's

Table 4: **Damage Taken (Absolute Values) — 90% DR Cap Applied**

Hit	**Baseline**	**Smith**	**Shaman (0 arm)**	**Shaman (25k arm)**	**Min HP: Smith**	**Min HP: S**
100	10	10	36	4	10	
500	83	50	181	18	50	
1,000	286	100	363	45	100	
2,000	889	333	725	146	333	
4,000	2,462	1,143	1,450	471	1,143	
8,000	6,095	3,556	2,900	1,387	3,556	
10,000	8,000	**5,000**	**3,625**	**1,923**	**5,001**	

Smith values at H=100, 500, and 1,000 corrected for the 90% DR cap (uncapped formula gave 1, 24, and 91 respectively — mechanically unachievable). Shaman (0 arm) damage = $H \times 0.3625$ *at all hit sizes. Shaman (25k arm) from per-element calculation with cap applied; matches the progression breakpoints table in Section 6. Smith outperforms 0-armour Shaman in absolute damage taken for* $H <$ *5,686; Shaman outperforms for* $H >$ *5,686.*

capped 90%. At H=1,000, the 250 physical residual lands at full value on a 0-armour Shaman; Smith absorbs 90% and takes only 100. For general mapping content where hit sizes stay below H≈2,000, Smith retains a real per-hit advantage over the zero-armour Shaman. *This is the window where Smith's investment pays off.*

The Crossover: Where Conversion Outperforms Armour

Smith's DR cap window closes at $H = 1{,}111$. Above this threshold, Smith's formula begins its decline:

$$\begin{aligned} H = 1{,}111&: \quad \text{Smith} = \frac{100{,}000}{111{,}110} = 90.0\% \quad \text{(cap boundary)} \\ H = 4{,}000&: \quad \text{Smith} = \frac{100{,}000}{140{,}000} = 71.4\% \\ H = 10{,}000&: \quad \text{Smith} = \frac{100{,}000}{200{,}000} = 50.0\% \end{aligned}$$

The Shaman (0 arm) remains at 63.75% throughout. The crossover — where Smith's declining DR curve meets Shaman's flat line — occurs at:

$$\frac{100{,}000}{100{,}000 + 10H} = 0.6375 \quad \Rightarrow \quad H = 5{,}686$$

Above H = 5,686: the 0-armour Shaman outperforms Smith of Kitava (100k armour). Smith has no mechanism to suppress H; every additional 1,000 raw damage adds 10,000 to the formula's denominator. Conversion intercepts this by reducing H at the source — the formula sees a 2,500 physical component instead of a 10,000 hit, and the remaining 75% never enters the formula at all.

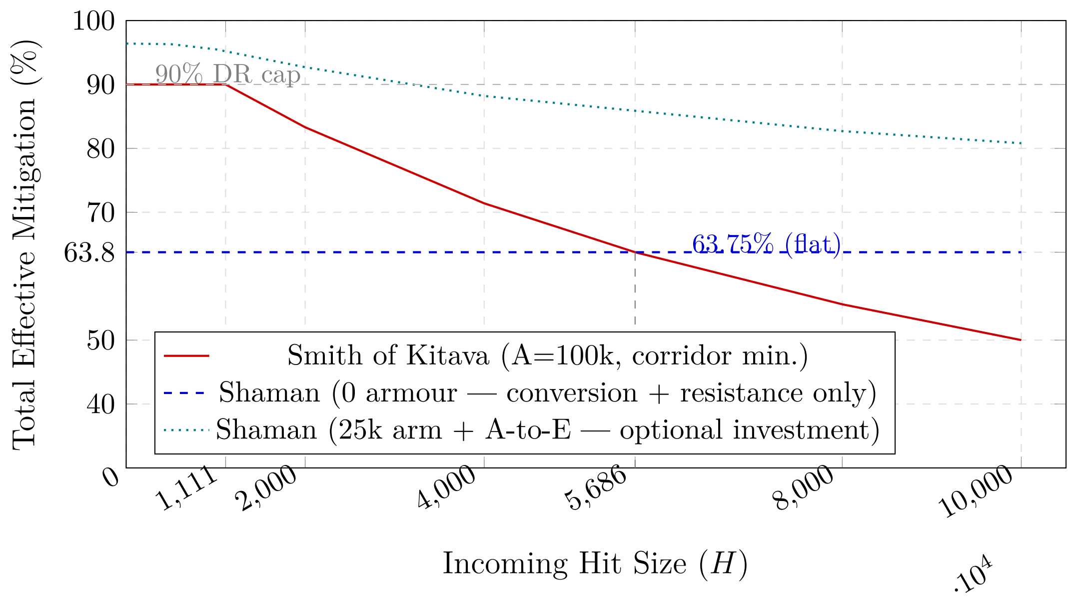

Figure 1: **Total Effective Mitigation vs. Hit Size — 90% DR Cap Applied.** Smith of Kitava (red solid): capped at 90% for $H \leq$ 1,111, then declining to 50% at the endgame ceiling. Shaman with no armour investment (blue dashed): a horizontal line at 63.75% — conversion and resistance are hit-size independent. Shaman with 25k armour and armour-to-elemental (teal dotted): a near-flat line from 96.4% to 80.8%, outperforming Smith at *every* hit size because resistance layers push composite mitigation above the 90% armour cap on small hits and conversion maintains efficiency on large hits. The crossover between Smith and 0-armour Shaman at H = 5,686 marks where the 0-armour conversion build begins outperforming the corridor-optimal armour build.

The figure captures the structural argument in one image. Smith's curve is a step function: horizontal at the 90% cap, then falling off a cliff at H = 1,111. Shaman's 0-armour line is horizontal throughout. The crossover is not a close call at the endgame scale — at H = 10,000, Smith is at 50% while Shaman (0 arm) is at 63.75%. All endgame boss strikes and map pinnacle encounters with genuine one-shot potential occur above H = 2,000; the majority of lethal content is above H = 5,000, which places it above the crossover. Smith's 90% capped advantage applies only to non-lethal chip damage.

The Armour Investment Decision

At $H = 10{,}000$, the 0-armour Shaman requires 3,625 effective HP to survive. The physical residual (2,500) reaches life without attenuation; the elemental components (1,125 at 85% resistance) are fixed by the resistance caps and do not depend on armour. Adding armour changes the physical component only:

$$\text{Physical saved by } A = 25{,}000\text{:} \quad 2{,}500 \times \frac{25{,}000}{50{,}000} = 1{,}250$$

The savings is **exactly 1,250 effective HP** — the physical component halved, independent of elemental resistance assumptions. Total damage falls to 2,375. The Shaman ascendancy's armour-to-elemental notables then extend armour coverage to each elemental component, reducing those from 1,125 to 673 and bringing the total to 1,923 — the full-stack figure used in the archetype sections.

Table 5: **Armour Investment Tiers — Shaman at H=10,000**

Configuration	**Physical taken**	**Elemental taken**	**Total taken**	**Min HP**
0 armour (baseline)	2,500	1,125	3,625	3,625
25k armour only (no A-to-E)	1,250	1,125	2,375	2,375
25k armour + 100% A-to-E (asc.)	1,250	673	1,923	1,923
Smith of Kitava (100k)	5,000	—	5,000	5,001

Armour-to-Elemental (A-to-E) is sourced from Shaman ascendancy notables, not passive tree nodes — it is available once those ascendancy slots are allocated. Adding 25,000 armour saves exactly 1,250 HP from the physical component at every hit size (always $0.5 \times 0.25H = 0.125H$). A-to-E then reduces elemental taken by a further 452 HP. Even the most conservative Shaman configuration (3,625 HP, 0 armour) requires 1,376 fewer effective HP than the minimum-viable Smith.

The investment question: is 1,250 effective HP or 25,000 armour the more accessible gain? For most Shaman builds, 1,250 effective HP can be sourced from life affixes on gear without entering armour cluster regions of the passive tree. Reaching 25,000 armour on a conversion build requires either armour-heavy gear rolls (competing with life and resistance affixes) or modest passive tree investment in armour clusters. Neither path is free; the relative cost depends on the specific build's passive tree position and gear budget.

The comparison with Smith clarifies the asymmetry. Smith must commit to *simultaneous* life and armour scaling constrained by the corridor: $A = 12\text{–}20 \times L$ means both resources must scale together. The Shaman decouples them: resistance handles the elemental majority, life handles the physical residual, and armour is optional scaling on top. Smith's binding constraint — that both A and L must reach minimums simultaneously — does not apply to a conversion archetype.

Quality of Life: The Case for Armour on Shaman

The ceiling analysis establishes that conversion and resistance alone outperform Smith at $H = 10{,}000$. However, the hit-size spectrum covers more than boss strikes. In mapping and general endgame content, the character absorbs continuous moderate hits in the $H = 100$–$H = 2{,}000$ range. At 0 armour, each hit delivers its physical residual at face value:

- $H = 100$: 25 physical + 11 elemental = 36 damage taken (36% of the hit reaches the life pool)
- $H = 1{,}000$: 250 physical + 113 elemental = 363 damage taken (36% of hit)

Neither value is a one-shot threat. But the consistent 36% bleed on moderate hits creates a continuous life-pool oscillation that degrades combat readability: the health globe fluctuates in proportion to incoming hit size rather than signalling genuine danger, and the physical residual is a raw number that scales linearly with map-content hit values. For players who track life pool depth as a real-time danger indicator, this is detectable noise.

Adding 25,000 armour collapses both values toward the 90% DR cap on the physical component. Because the Shaman's physical component is only 25% of the original hit, the armour-to-physical crossover into the capped regime happens at the same original hit size as Smith: $H \leq 1{,}111$. Above that threshold, armour's DR on the physical residual still dramatically reduces it:

- $H = 100$: physical DR = 90% (capped) $\rightarrow$ 2.5 physical + $\approx$1 elemental $\approx$ 4 total (96% composite)
- $H = 1{,}000$: physical DR = 90% (capped) $\rightarrow$ 25 physical + $\approx$20 elemental $\approx$ 45 total (95.5% composite)

The life-pool oscillation is reduced to noise. The health globe reads as stable across mapping content unless a genuinely dangerous hit arrives. Whether this quality-of-life gain justifies the 25,000 armour investment — and the 1,250 HP reduction in the ceiling-survival floor it trades against acquiring additional HP — is a player-preference decision, not a survivability argument. Players who value ceiling performance over daily-play smoothness may prefer the pure HP path; players who prefer a stable life globe in mapping should consider the armour investment even if pure HP is mathematically equivalent at the endgame ceiling.

Minimum Survival at the Endgame Hit Ceiling

Table 6: **Survival Analysis — 10,000 Raw Physical Hit (Endgame Ceiling)**

Build	**Damage Taken**	**Min HP to Survive**	**Survives at 3,625 H**
Baseline (25k arm.)	8,000	8,001	
Smith of Kitava (100k)	5,000	5,001	
Shaman (0 arm, 75% conv.)	**3,625**	**3,625**	**Y**
Shaman (25k arm, full stack)	1,923	1,924	Yes (comfortab
Shaman (min-maxed)	1,388	1,389	Yes (ample marg

Reference pool set to 3,625 HP — the minimum for 0-armour Shaman survival at the endgame ceiling. Smith at 100k armour cannot survive this hit at that pool; it requires 5,001 HP. Shaman with 25k armour and full ascendancy stack produces 1,702 HP of margin at the same reference pool. Min-maxed Shaman assumes 35,000 armour and 90% max resistance.

The Mind over Matter Compound

In Path of Exile 2, Mind over Matter (MOM) operates as a full pre-life shield: **100% of incoming damage is taken from the mana pool before any reaches life**. All hit damage depletes mana first; only the portion that exceeds the mana pool overflows to life. The eHP pool for single-hit survival is therefore fully additive:

$$\text{Effective eHP (full MOM)} = \text{Life} + \text{Mana}$$

Mana is not a percentage reducer — it is a second life pool. The survival requirement does not change; the character sheet now has two pools that jointly cover it.

The partial variant — passives granting "10% of damage taken from mana before life" — routes only 10% of each hit to mana and 90% to life. Mana depletes at a tenth the rate, preserving skill-casting ability far longer. If the mana pool holds at least $L/9$ (where L is the life pool), the effective HP the life pool can absorb becomes:

$$\text{Effective eHP (10\% variant)} = \frac{L}{0.9} \qquad \text{(minimum mana: } L/9)$$

The core eHP requirement from the survival tables in this section does not change under either variant. What changes is which character-sheet resources cover that requirement, and the operational risk profile of each path.

Table 7: **MOM eHP — Shaman Tiers vs. Smith at H=10,000**

Build	**Damage**	**MOM type**	**Effective eHP**	**Min mana neede**
Shaman (0 arm)	3,625	None	Life $\geq$ 3,625	0
Shaman (0 arm)	3,625	10% partial	Life /0.9 $\geq$ 3,625	$L/9 \approx 363$
Shaman (0 arm)	3,625	Full MOM	Life + Mana $\geq$ 3,625	any split $\geq$ 3,625
Shaman (25k arm, full stack)	1,923	Full MOM	Life + Mana $\geq$ 1,923	any split $\geq$ 1,923
Smith of Kitava (100k)	5,000	Full MOM	Life + Mana $\geq$ 5,001	any split $\geq$ 5,001

Full MOM makes life and mana fully fungible for single-hit survival: any combination summing to the required total survives the hit. The 10% partial variant requires only a small mana pool ($L/9$) to fully activate its 1.11× life multiplier. Neither variant reduces the damage that must be absorbed — both shift which resources absorb it. Full MOM drawbacks: mana depletion prevents skill casting, halves mana regeneration rate, and requires mana flask recovery between threatening encounters; a second large hit before mana restores hits the life pool at full value.

The comparative advantage of MOM on Shaman vs. Smith is not a compounding of conversion efficiency — it is that Shaman arrives with a lower damage requirement, so the mana (or life) budget needed to cover the MOM pool is proportionally smaller. A 0-armour Shaman pursuing full MOM needs a combined pool of 3,625 from any mix of life and mana. Smith needs 5,001 — and must build that pool on top of 100,000 armour, compressing the character-sheet budget across three resources simultaneously. The risk profile of a depleted mana pool is also worse on Smith: mana recovery returns 5,000 eHP to Smith's shield; the same event returns only 3,625 on Shaman, but the Shaman's life pool is the entire floor during recovery while Smith's is capped at 3,001 (if split evenly).

4 Witchhunter: Sorcery Ward and the Absorption-Layer Paradigm

Class Overview and Defensive Paradigm

The Witchhunter is an ascendancy of the Mercenary base class and represents the third distinct defensive architecture examined in this analysis. Where Shaman and Infernalist achieve mitigation by suppressing the effective hit size through physical-to-elemental conversion, and where the Invoker achieves efficiency through avoidance, the Witchhunter sidesteps the armour formula's structural weakness—its quadratic degradation against large hits—through a fundamentally different mechanism: an absorption layer that intercepts post-armour damage before it reaches the life pool.

This distinction is categorical. Damage reduction (DR) scales as $A/(A + 10H)$, which collapses toward zero as H grows regardless of armour investment. An absorption pool absorbs a flat quantum of post-DR damage without degrading with hit size. Against very large hits, the Witchhunter's Sorcery Ward (SW) therefore provides more reliable protection per point of investment than additional armour provides. The Witchhunter achieves this without a unique-item dependency, making its defensive model fully accessible in Solo Self-Found, Hardcore, and HC-SSF contexts.

The Two Mandatory Ascendancy Nodes

The Witchhunter's absorption stack requires two of its four ascendancy notables. These are non-negotiable; the remaining two slots are available for offensive or utility investment.

Sorcery Ward Mechanics

Sorcery Ward operates as an absorption layer placed between the armour DR step and the life pool in the damage sequence:

$$\text{SW Pool} = \min\Big(32{,}000,\ 0.30 \times (\text{Post-Penalty Armour} + \text{Post-Penalty Evasion})\Big)$$

$$\text{Effective Armour (post-penalty)} = \text{Geared Armour} \times 0.50$$

Table 8: **Witchhunter Ascendancy Nodes — Defensive Core**

Node	**Effect**	**Priority**
Obsessive Rituals	50% less to armour and evasion ratings. Grants Sorcery Ward: absorption pool equal to 30% of post-penalty combined armour+evasion, capped at 32,000.	Mandatory
Ceremonial Ablution	Extends SW coverage from elemental hit damage to *all* hit damage types, including physical and chaos.	Mandatory
Third/Fourth Choice	Offensive scaling, utility, or situational defensive investment.	Situational

Obsessive Rituals is the gating node for the SW mechanic; Ceremonial Ablution converts it from an elemental-only to an all-damage absorption pool. Both must be taken to achieve full-damage-type coverage. The 50% less multiplier on armour and evasion is a final multiplier, not overcome by increased armour nodes; effective armour post-notables is exactly half of geared armour.

Damage Sequence: Armour DR $\rightarrow$ SW absorbs remainder $\rightarrow$ Life receives overflow

SW recharges fully on a 6-second cooldown. Once depleted by a hit, SW is unavailable until the cooldown expires — making the build vulnerable to rapid burst clusters that deliver multiple large hits within the 6-second window.

At 60,000 geared armour: post-penalty effective armour = 30,000. Sorcery Ward $= 0.30 \times 30{,}000 = 9{,}000$. At $L = 3{,}000$ HP, total absorption stack $= 9{,}000 + 3{,}000 = 12{,}000$.

Reaching the 32,000 SW cap requires a post-penalty armour+evasion pool of approximately 107,000, or approximately 213,000 before the 50% penalty — confirming that 9,000 is a realistic and achievable mid-endgame figure.

Hit-Size Survival Table

The following table applies $\text{DR} = A/(A + 10H)$ at the post-penalty armour of 30,000, then deducts the post-DR remainder from SW = 9,000, with overflow reaching life at $L = 3{,}000$.

Table 9: **Witchhunter Survival — 60,000 Geared Armour (30,000 Eff.), SW = 9,000, L = 3,000 HP**

Raw Hit	DR	Post-DR	SW Absorbs	Life Dmg	Survives?
5,000	37.5%	3,125	3,125	0	Yes
8,000	27.3%	5,818	5,818	0	Yes
10,000	23.1%	7,692	7,692	0	Yes
12,000	20.0%	9,600	9,000	600	Yes
14,000	17.6%	11,529	9,000	2,529	Yes
14,485	**17.1%**	**12,007**	**9,000**	**3,007**	**Threshold**

A = 30,000 (post 50% less), SW = 9,000, L = 3,000. Survival ceiling is derived from setting life damage ≤ L: $10H^2/(30{,}000 + 10H) - 9{,}000 = 3{,}000$*, which solves to* $H \approx 14{,}485$*. Against the same 10,000 raw hit at equal investment: Smith at 60,000 geared armour and L = 3,000 takes approximately 3,509 (exceeding L, structurally fatal at this pool). Shaman at zero armour and L = 3,000 takes approximately 3,175 (also exceeding L without additional Energy Shield). Witchhunter takes **zero life damage** from the same hit and survives with full life intact. This is the clearest expression of the absorption paradigm's ceiling advantage.*

The survival ceiling of approximately $H = 14{,}485$ against a raw physical hit substantially exceeds what Shaman or Smith can achieve at comparable investment and life pool. The Witchhunter's SW pool achieves this without any unique-item requirement, passive-point over-investment, or Energy Shield stacking — purely from two ascendancy notables applied to the same armour investment other builds treat as a DR source.

Cross-Damage-Type Coverage

Ceremonial Ablution extends SW coverage to physical, elemental, and chaos hit damage simultaneously. This is structurally superior to the other archetypes' chaos defences:

- **Shaman:** No chaos conversion or absorption layer; chaos damage lands with only resistance mitigation. Chaos is Shaman's exposed damage type.
- **Smith of Kitava:** Dedication to Kitava applies armour DR to chaos hits — effective but limited by the same $A/(A + 10H)$ collapse at large hits.
- **Infernalist/Invoker:** Chaos Inoculation provides complete immunity to chaos damage at the cost of life = 1.
- **Witchhunter:** SW intercepts post-armour chaos damage using the same pool

and the same damage sequence. Against a 10,000 raw chaos hit at $A = 30{,}000$: DR = 23.1%, post-DR = 7,692, SW absorbs 7,692 — zero life damage. Coverage is equivalent to the physical case shown above.

The Witchhunter is the only archetype in this paper that achieves baseline absorption against physical, elemental, and chaos hit damage from a single two-notable investment, without either requiring a unique item (Shaman: Cloak of Flame) or committing the entire life pool to Energy Shield (CI builds).

Evasion Interaction and the Armour-Evasion Hybrid

Sorcery Ward scales with the *combined* post-penalty armour and evasion pool, making evasion investment doubly efficient for the Witchhunter:

1. Each point of post-penalty evasion contributes 30% of its value to SW, expanding the absorption pool.
2. Evasion provides a hit-avoidance layer that reduces the rate at which the SW pool is consumed per encounter.

An armour/evasion hybrid Witchhunter running a shield achieves the post-penalty armour floor at lower geared investment while simultaneously expanding SW from the evasion contribution. At 50% evasion chance and 50% block chance, 75% of attacks are avoided before the SW pool is engaged — meaning SW is consumed at one-quarter the rate of a pure armour build facing the same encounter. This makes the 6-second recharge cooldown substantially less punishing in practice.

Weapon-Swap Passive Efficiency

Path of Exile 2 allows passive nodes to be allocated to a specific weapon set, becoming active only while that set is equipped. For the Witchhunter, this creates a structural passive-point surplus: nodes on the inactive weapon set cost no investment during the active set's operation.

In practice, a secondary weapon set carries defensive or utility nodes — armour scaling, resistance nodes, SW-contributing clusters — while the primary set runs an entirely offensive allocation. The effective passive budget available to the character exceeds the nominal count, because neither set's nodes compete with the other while they are not simultaneously active.

Two utility nodes are particularly well-suited to the secondary weapon set:

- **Culling Strike:** Kills enemies below a life threshold (typically 10% maximum life). Against enemies that would otherwise require two hits, culling compresses two engagements into one, adding approximately 11% kill efficiency across a full pack. On boss encounters with regenerating life, the execute eliminates the window during which damage must outpace recovery.
- **Explode:** Enemies killed explode dealing a percentage of their maximum life as area physical damage, converting every kill into a secondary clear event. Against dense endgame monster packs, a single killed enemy propagates damage across the remaining pack without an additional skill cast.

Allocated to the secondary weapon set, both nodes cost zero primary-set passive investment; the player retains the full primary-set budget for damage and survivability nodes while gaining substantial clear-speed upside on demand.

Trade-offs

- **Burst vulnerability:** SW recharges on a 6-second cooldown. Rapid consecutive large hits deplete SW on the first strike and leave life exposed until the cooldown completes. Unlike Invoker's Meditate (pre-combat reset) or Shaman's passive DR (always active), SW cannot be refreshed mid-encounter on demand.
- **Damage over time bypass:** Bleed, poison, ignite, and other DoT effects bypass SW and must be managed through flask charges or dedicated mechanics. SW only intercepts hit damage.
- **50% less armour:** The Obsessive Rituals penalty reduces effective armour to half of geared investment. The standard 12×–20× armour optimality corridor implies a corresponding increase in geared armour targets. However, each additional point of geared armour simultaneously improves DR and expands SW — making armour investment more efficient per passive point for a Witchhunter than for a zero-SW build at the same armour tier.

Witchhunter Positioning

Table 10: **Witchhunter vs. Shaman vs. Invoker: Defensive Profile Comparison**

Property	**Shaman**	**Witchhunter**	**Invoker**
Primary defence	Conversion (DR suppression)	Sorcery Ward (absorption)	Avoidance (evasion +
DR on landed H=10k	80.8% (all layers)	23.1% (armour only)	27.3% (50k evs)
Post-DR absorption	None	9,000 SW (mid-endgame)	None (Meditate pre-co
Survival: H=10k	1,923 HP needed	0 life damage (SW covers)	4,849 ES + Meditate
Chaos damage	Near-zero residual	SW absorbs post-DR	CI immunity (keyston
eHP source	Life (conventional)	Life + SW buffer	Energy Shield (CI)
Unique item required	Yes (Cloak of Flame)	No	No
Playstyle constraint	None	SW cooldown awareness	Meditate pre-combat
Survival ceiling (H)	≈10,000 (standard)	≈14,485 (mid-invest)	Conditional (Meditate
Verdict	**Point-efficient, passive**	**Highest one-shot ceiling**	**Encounter-level avo**

Witchhunter's survival ceiling is the highest of the three at comparable passive investment, achieved without unique-item dependency. The cost is cooldown vulnerability on burst damage sequences and DoT exposure. Shaman's profile is passive and consistent; Invoker's is avoidance-primary with a conditional eHP ceiling via Meditate; Witchhunter's is absorption-primary with the highest sustainable hit ceiling when SW is charged. Each archetype answers a different threat profile, and each is viable; none is universally superior.

5 Infernalist: Chaos Inoculation and the 20% Hard Immunity

Class Overview

The Infernalist is an ascendancy of the Witch base class and occupies a unique categorical position in the conversion hierarchy. Like Shaman, it accesses physical-to-other-damage-taken conversion through an ascendancy notable — but the target element is chaos rather than elemental, and the interaction with the Keystone Passive Skill **Chaos Inoculation (CI)** transforms what would otherwise be a near-zero elemental reduction into a categorical hard immunity. This distinction, examined precisely, separates Infernalist from every other defensive archetype discussed in this document.

The Infernalist's ascendancy notable *Altered Flesh* converts 20% of all physical damage taken to chaos damage. Combined with Cloak of Flame (50% physical to fire), the baseline conversion stack reaches 70%, leaving 30% residual physical. With CI active, the 20% chaos portion ceases to interact with armour or resistance — it simply does not happen. The build is immune to it.

Conversion Stack: Two Configurations

Table 11: **Infernalist Conversion Stack — Baseline vs. Full Optional Investment**

Source	Type	Element	Baseline	Full (Optional)
Cloak of Flame	Unique Chest	Fire	50%	50%
Molten Being	Passive Tree Notable	Fire	—	5%
Altered Flesh	Ascendancy Notable	Chaos	20%	20%
Catalysis	Anointment	Random Element	—	5%
Total Converted			**70%**	**80%**
Residual Physical			**30%**	**20%**

Molten Being is located on the northwest side of the passive tree and requires non-trivial passive investment to reach. Catalysis is an anointment consuming the amulet slot. The Full configuration commits both resources, raising fire conversion to 55% and total conversion to 80%. The chaos component (20%) is reduced to 0 under Chaos Inoculation regardless of configuration — it does not enter any further calculation. Fire sub-total in Full = 50 + 5 = 55%; chaos = 20% (immune); random = 5%; residual physical = 20%.

The Categorical Distinction: Hard Immunity vs. Near-Zero Reduction

All previous builds in this document reduce elemental contributions to near-zero through sequential layers (armour-to-elemental, resistance, less-elemental-taken). "Near-zero" is not zero. At H=10,000, Shaman's cold, lightning, and random elemental components each resolve to approximately 12 damage after all layers — small, but nonzero, and the total elemental contribution still represents 671 damage before the "less elemental taken" multiplier.

The Infernalist with CI operates differently on the chaos component. The 20% of the hit that converts to chaos does not pass through any mitigation layer. It is avoided in its entirety:

$$\text{Chaos damage taken} = 0.20 \times H \times (1 - \text{CI immunity}) = 0.20 \times H \times 0 = 0$$

At H=10,000, CI avoids **exactly 2,000 damage** — hard, not as a limit approached asymptotically. This is also the most reliable defensive statement possible: chaos immunity does not degrade with hit size, does not depend on armour values or resistance caps, and cannot be bypassed by resistance penetration or reduction. The 20% disappears categorically regardless of what the hit does.

This also addresses chaos damage from other sources entirely. In current Path of Exile 2 endgame content, chaos damage is among the most lethal because it bypasses most conventional mitigation. CI converts this from the build's most dangerous threat into its most completely nullified one.

Hit-Size Comparison: Infernalist vs. Shaman

The following table uses the same build parameters as prior sections (25,000 armour, 100% armour-to-elemental, 85% max resistance), with CI accounting for the full 20% chaos exemption.

Table 12: **Damage Taken and Mitigation — Shaman vs. Infernalist Configurations**

Hit	Shaman Tkn	Sham%	Inf. Base Tkn	IB%	Inf. Full Tkn	IF%
100	<1	99.5%	<1	99.5%	<1	99.7%
500	11	97.8%	12	97.6%	8	98.4%
1,000	41	95.9%	45	95.5%	30	97.0%
2,000	143	92.8%	159	92.1%	106	94.7%
4,000	469	88.3%	523	86.9%	351	91.2%
8,000	1,387	82.7%	1,545	80.7%	1,054	86.8%
10,000	1,923	80.8%	2,136	78.6%	**1,469**	**85.3%**

"Inf. Base" = Infernalist baseline: 50% fire, 20% chaos (CI = 0), 30% physical. "Inf. Full" = Infernalist full: 55% fire, 20% chaos (CI = 0), 5% random, 20% physical. Shaman column reproduced from Section 3 for reference (25k armour configuration). All configurations use 25,000 armour, 100% armour-to-elemental, 85% max elemental resistance. For consistency with the 0-armour Shaman baseline established in Section 3: 0-armour Infernalist Baseline takes 3,750 at H=10,000 (3,000 physical + 750 fire after 85% res; CI eliminates the 20% chaos component regardless of armour); 0-armour Infernalist Full takes 2,900 (2,000 physical + 825 fire + 75 random). Infernalist Full outperforms the 0-armour Shaman (3,625) even at zero armour, because CI eliminates 2,000 damage categorically and the reduced physical residual (20%) keeps the formula's denominator small. Infernalist Baseline (3,750) does not: the 30% physical residual costs more than CI recovers at 0 armour. Note that the Infernalist Baseline takes more damage than the 25k-armour Shaman at all hit sizes despite CI, because the larger physical residual (30% vs 25%) more than offsets the chaos exemption at this armour level. The Full configuration resolves this and outperforms Shaman at every hit size.

Why the Baseline Underperforms Shaman Despite CI

At first glance, negating 20% of a 10,000 hit (2,000 damage hard avoided) should place Infernalist ahead of Shaman. The number shows the opposite: Infernalist Baseline takes 2,136 versus Shaman's 1,922. The reason is the physical residual:

- Shaman: 25% residual physical (2,500 effective hit) → armour DR = 50% → 1,250 taken
- Infernalist Base: 30% residual physical (3,000 effective hit) → armour DR = 45.5% → 1,636 taken

The 386-unit increase in physical damage taken (1,636 vs 1,250) exceeds the 500-unit reduction from the CI fire-contribution shift. CI is extremely efficient at avoiding the chaos component — but the baseline conversion stack leaves enough physical residual that the armour formula's larger denominator costs more than CI saves, at this armour value.

This resolves cleanly when the Full optional stack is active. Reducing physical residual to 20% (2,000 effective hit) gives 55.6% armour DR and 889 damage taken — well below Shaman's physical component — producing the 1,469 total that outperforms Shaman.

Accelerating Returns on Conversion: The Scaling Efficiency Argument

Each additional 5 percentage points of physical-to-other conversion is worth *more* than the previous 5 percentage points. This is a direct consequence of the fact that the gain is calculated against a shrinking denominator (the remaining physical residual):

$$\text{Gain of next 5pp} = \frac{\Delta\text{Residual}}{\text{Residual Before}} = \frac{5\%}{\text{Current Residual}}$$

This table validates the user's intuition precisely. The Shaman's 70% → 75% step was identified as a "monumental leap" — 16.7% less physical damage per hit. The Infernalist Full's 75% → 80% step is worth 20.0%, 20% more gain than the Shaman's step, confirming that it represents a strictly greater efficiency increment, not merely an equivalent one.

Table 13: **Accelerating Returns: Relative Value of Each 5 pp Conversion Increment**

Conversion Step	**Residual Before**	**Residual After**	**Physical**
60% → 65%	40%	35%	12.5% les
65% → 70%	35%	30%	14.3% les
70% → 75%	**30%**	**25%**	**16.7% (**
75% → 80%	**25%**	**20%**	**20.0% (Inferna**
80% → 85%	20%	15%	25.0% les
85% → 90%	15%	10%	33.3% les
90% → 95%	10%	5%	50.0% les
Key Insight	**Each 5 pp step is more valuable than the last — gains accelerate as c**		

"Physical Reduction Gain" is the percentage reduction in residual physical damage per hit relative to the previous step. This is not a marginal effect — moving from 75% to 80% conversion (Infernalist Full's contribution) produces 20% less physical damage taken per hit, compared to 16.7% for the Shaman's 70% → 75% step. The same logic applies in both directions: viewed in reverse, failing to acquire the next 5 pp of conversion costs increasingly more as you approach high total conversion. The Infernalist Full's 75% → 80% step is categorically more valuable than the Shaman's equivalent preceding step.

Opportunity Cost: Molten Being vs. Energy Shield

The Full configuration requires Molten Being (passive tree notable, non-trivial pathing cost) and Catalysis (anointment slot, consuming the amulet annoint). The user correctly frames this as an opportunity cost decision rather than a clear win:

- **Passive points for Molten Being** could alternatively deepen the Energy Shield pool, increasing the effective HP ceiling for CI. A larger ES pool survives larger hits before the mitigation percentage even applies.
- **The anointment slot** is consumed by Catalysis; other anointments providing ES, resistance, or damage may provide greater net value depending on the build's current gaps.
- **However:** The 20% reduction in physical residual from Infernalist Full vs. baseline (shown in the hit-size comparison table above) reduces the survival HP requirement from 2,137 to 1,469 — a 668 HP difference at H=10,000. Whether 668 ES from the anointment alternative exceeds this survival margin depends entirely on build-specific ES totals.

At moderate ES pools (<4,000), the Full configuration likely provides a superior marginal survival guarantee per passive point than additional ES would. At very large

ES pools (>6,000), the baseline configuration's survival floor is already comfortable, and the Molten Being path may be less efficient than ES amplification. The decision is build-stage dependent.

Infernalist Positioning

Table 14: **Infernalist Defensive Profile vs. Shaman at H=10,000**

Configuration	Conv.	Damage Taken	Mitigation	Min HP
Shaman (standard)	75%	1,923	80.8%	1,923
Infernalist Baseline (CI)	70% + CI	2,136	78.6%	2,136
Infernalist Full (CI)	**80% + CI**	**1,469**	**85.3%**	**1,469**
Shaman Min-Maxed	75%	1,388	86.1%	1,388
Infernalist Full Min-Maxed	80% + CI	est. <1,100	est. >89%	est. <1,100

All entries at 25,000 armour, 100% armour-to-elemental, 85% max resistance unless noted. Min-Maxed Shaman uses 35,000 armour and 90% max resistance. Infernalist Full Min-Maxed is estimated, incorporating higher armour (35,000), 90% max resistance, and the "less elemental taken" multiplicative layer; CI immunity on the chaos component is maintained regardless of other parameters. The Infernalist Full at standard investment surpasses the Shaman at standard investment and approaches the Shaman Min-Maxed ceiling — while being supported by an ES pool (CI) rather than a conventional life pool, and providing categorical chaos immunity as an additional defensive property.

The Infernalist represents a genuine alternative defensive powerhouse rather than a lateral trade. Its distinguishing properties relative to Shaman are: (1) a hard chaos immunity rather than a near-zero residual on the converted component, (2) an ES-only effective HP pool under CI that interacts favorably with Energy Shield mechanics, and (3) access to spell-casting, minion, and select martial weapon playstyles, with the ascendancy favouring fire and spell-caster builds most heavily. The baseline configuration is slightly below Shaman in raw physical mitigation but offers chaos immunity as compensation; the Full configuration exceeds Shaman in raw physical mitigation while retaining that immunity, at the cost of committed passive investment.

6 Shaman: Physical-to-Elemental Conversion and the Gold Standard

Conversion Stack: Sources and Assembly

The Shaman's physical-to-elemental conversion assembles from six distinct sources: a unique chest item, three ascendancy notables, a passive tree notable, and a crafted anointment. Each source stacks additively.

Table 15: **Shaman Physical-to-Elemental Conversion Stack — All Sources**

Source	Type	Target Element	Conversion
Cloak of Flame	Unique Chest Item	Fire	50%
Ascendancy Notable	Ascendancy	Fire	5%
Passive Tree Notable	Passive Skill Node	Fire	**5%**
Ascendancy Notable	Ascendancy	Cold	5%
Ascendancy Notable	Ascendancy	Lightning	5%
Anointment	Ring / Amulet Craft	Random Element	5%
Total Elemental Converted			**75%**
Residual Physical			**25%**

Note: Fire receives 60% in total (50 + 5 + 5). Cold, lightning, and the random anointment element each receive 5%. The bolded passive tree notable (5% physical taken as fire) is the corrected source that prior analyses omitted, raising the aggregate from 60–70% to 75% and changing residual physical from 40% to 25%.

The correction from prior analysis matters nonlinearly. Moving from 60% to 75% conversion reduces residual physical from 40% to 25% of the original hit — a 37.5% reduction in the physical component that the armour formula must handle. Because armour's efficiency scales with *how small* the incoming hit is (not linearly with conversion), this final increment of conversion produces disproportionately large gains, as the hit-size tables in Section 3 demonstrated.

Full Defensive Layer Synergy

The hit-size tables in Section 3 use conservative assumptions: they include conversion, armour, armour-to-elemental, and resistance, but omit the multiplicative "less elemental damage taken" ascendancy notable and block. Both of these layers compress the final taken number further.

Table 16: **Shaman Defensive Layer Summary — All Layers**

Layer	Function	Shaman Access
Phys-to-Ele Conversion	Reduces effective physical hit size; feeds elemental layers	75% via Cloak + ascendancy + tree notable + anoint
Armour (physical DR)	Mitigates the 25% residual physical hit	50% DR at 25k armour vs 2,500 effective hit
Armour-to-Elemental	Extends armour formula to each converted ele hit independently	100%+ via ascendancy nodes
Max Elemental Resistance	Caps each element type at 85–90%	Primal Aegis ascendancy path
Less Elemental Taken	Multiplicative reduction after resistance — stacks with all above	Dedicated ascendancy notable
Block	Full hit avoidance before all mitigation layers	At or near maximum cap via passive placement
Hybrid eHP (Life / Mana / ES)	Expands effective health pool beyond raw life	MOM + EB accessible on Shaman passive pathing
Combined	**Elemental component near-zero; physical DR 4× peers**	**All layers within normal progression budget**

MOM = Mind over Matter; EB = Eldritch Battery. The "less elemental taken" multiplier is not included in the Section 3 tables; it provides an additional 15–25% multiplicative reduction on the fire component (635 damage at H=10,000), pushing total mitigation from 80.8% to an estimated 82–85%. Min-maxed configuration with this layer would approach 87–89% total mitigation at the endgame ceiling.

The Elemental Collapse in Detail

At H=10,000, the converted elemental portion (7,500 raw) is handled across three sequential reduction layers:

1. **Armour-to-Elemental:** The 6,000 fire hit faces 29.4% DR from 25,000 armour, leaving 4,235. Each 500-damage cold/lightning/random hit faces 83.3% DR, leaving only 83 damage each.
2. **Max Resistance (85%):** The 4,235 post-armour fire becomes 635. The 83 per-element remnants become 12 each (36 total for three components).
3. **Less Elemental Taken (est. 20% multiplier):** Compresses the 671 combined elemental to approximately 537.

Against the 25% residual physical (1,250 taken), the elemental component is now less than half of the physical component. The "collapse to near zero" is not hyperbole — 7,500 of converted elemental damage becomes roughly 537 final damage through three sequential reduction layers.

Passive Point Economy

Shaman's mitigation superiority is delivered at nearly zero passive budget cost, creating a compound advantage:

1. **Freed points target damage.** The 8–15 passive points Smith commits to armour% clusters become damage nodes, jewel sockets, or life/mana scaling on Shaman. This is not a trade-off — it is a unilateral gain, because the freed points come at no defensive cost (Shaman is already more defensive than Smith).
2. **Gear affixes reallocate.** Armour% suffixes Smith requires become resistance, life, or damage affixes on Shaman. Conversion through ascendancy and a chest item does the work that multiple gear suffixes would otherwise need to cover.
3. **No defensive floor sacrifice.** Because Shaman's mitigation exceeds Smith's at every hit size above roughly H=1,000 (the endgame-relevant range), there is no scenario where redirecting passive points to damage creates a defensive shortfall relative to the competing archetype.

Shaman can simultaneously out-defend a committed armour archetype and out-damage (or out-sustain) it through the budget reclaimed from the armour tree.

Progressive Investment Breakpoints

Table 17: **Shaman Defensive Benchmarks by Progression Stage — H=10,000 Ceiling**

Stage	Conv.	Armour	A-to-E	Max Res	Taken	Mitigation
0-Armour Baseline	75%	0	0%	85%	3,625	63.75%
Early Maps	50%	15,000	50%	75%	4,934	50.7%
Mid Red Maps	60%	20,000	75%	80%	3,627	63.7%
Late Endgame	75%	25,000	100%	85%	1,923	80.8%
Min-Maxed	75%	35,000	100%	90%	1,388	86.1%

0-Armour Baseline: full 75% conversion stack assembled (Cloak of Flame + all ascendancy notables + passive tree notable + anointment) with zero armour investment; represents the theoretical floor before any armour decision is made. Early Maps assumes Cloak of Flame equipped but partial ascendancy investment; no passive tree notable or anointment yet; 63.75% flat mitigation beats Early Maps (50.7%) because Early Maps has reduced conversion and lower resistance caps. Min-maxed does not include the "less elemental taken" multiplicative layer, which would push effective mitigation to approximately 87–89%.

The breakpoint table illustrates a key property of this configuration: the Shaman is not a late-game gimmick. Even with Cloak of Flame alone and no other conversion source (Early Maps), the build takes 4,934 damage from a 10,000 hit — equivalent to a Baseline build with roughly 40,000 armour, without having invested a single passive point in armour scaling.

Shaman Positioning

Shaman's S-Tier classification is a claim about floor, not ceiling. The Witchhunter's absorption ceiling ($H \approx$ 14,485) is higher; Infernalist Full's VHP multiplier (6.8×) is higher; Warbringer's expected damage per hit (1,055) is lower. Shaman's defining property is that none of its performance is conditional. Every hit, every encounter, every probability roll: the build delivers 80.8% total mitigation at standard investment with no skill activation, no cooldown, and no unique item dependency beyond a single chest slot. The three Degenerate Gamblers that follow are not weaker archetypes — their expected performance is competitive. They carry variance Shaman does not.

Table 18: **Shaman vs. Degenerate Gamblers at H=10,000**

Property	**Shaman**	**Invoker**	**Warbringer**
Tier	S (Deterministic)	A (Gambler)	A (Gambler)
VHP/eHP	5.2×	>5× encounter	6.6× exp / 2.0× wc
Expected damage	1,923	1,818 (25% land)	1,055
Worst-case damage	1,923	7,270 (no Meditate)	3,556
Variance	Zero (deterministic)	High	Moderate
Skill activation	No	Meditate (pre-combat)	Manual block
Min HP, worst case	1,924	≈5,000 ES	3,557
Unique item required	Yes (Cloak of Flame)	No	No
Verdict	**S-Tier floor, passive**	**Encounter-efficient, bursty**	**Best expected; varianc**

Shaman figures (1,923 / 1,923) assume full-stack configuration: 75% conversion, 25,000 armour, 100% armour-to-elemental, 85% max resistance. The 0-armour Shaman baseline (Section 3) takes 3,625 at H=10,000 — deterministically lower than Smith (5,000), lower than Invoker worst-case (7,270), and comparable to Warbringer worst-case (3,556); all HP requirements in this table must be met for the full-stack configuration shown. Invoker expected damage: 25% of hits land at 7,270 each (27.3% DR at 50k evasion); expected per hit = 1,818. Warbringer and Tactician figures from their respective positioning sections. Shaman's S-Tier placement rests on determinism: its worst case equals its expected case because conversion is unconditional. Each Gambler's expected mitigation is competitive with or exceeds Shaman's, but their worst-case damage demands 1.8–3.8× the HP investment the full-stack Shaman requires. The skill floor is higher on each Gambler — Meditate for Invoker, manual block for Warbringer, totem management for Tactician — and variance rises further when those mechanics are unavailable, mistimed, or already spent.

The Degenerate Gamblers

The three archetypes in this section share a defining structural property: their defensive outcome on any given hit is determined by a probability chain rather than a guaranteed calculation. Invoker, Warbringer, and Tactician each operate through a primary avoidance or reduction roll — evasion, block, or deflection — that either fires or does not. When it fires, the result is exceptional. When it does not, the character absorbs damage at or near the armour-pure floor.

This is the **Degenerate Gambler** archetype family: high skill floor, high skill expression, and high variance. Each provides active mechanics that allow a skilled player to convert probabilistic outcomes into deterministic ones at key moments — *Meditate* before an encounter for Invoker, manual block on telegraphed slams for Warbringer, and positioning with totem placement for Tactician. Under optimal play, the ceiling of each archetype is competitive with the S-Tier deterministic specialists. Under worst-case probability rolls, the floor is not.

The critical observation: even at worst case, all three Degenerate Gambler archetypes outperform the Armour Pure baseline (2.0× VHP/eHP). This is not a consolation statistic — it is a hard architectural claim. A Warbringer who fails to block, an Invoker who takes a landed hit without Meditate, or a Tactician whose deflection roll fails still carries a better defensive profile than a character who invested exclusively in raw armour and life. The gamble is not whether to play a Gambler archetype versus a pure archetype. The gamble is whether the player can push the probability chain toward the favourable outcome consistently enough to compete with S-Tier.

7 Invoker: Avoidance, Evasion-as-Armour, and the Conditional eHP Ceiling

Class Overview and Defensive Paradigm

The Invoker is an ascendancy of the Monk base class and represents a categorically different defensive architecture from the conversion builds documented in preceding sections. Where Shaman and Infernalist achieve passive per-hit mitigation by suppressing the effective hit size through physical-to-elemental conversion, the Invoker achieves defensive efficiency through *avoidance* — preventing hits from connecting at all — backed by a novel ascendancy mechanic that converts evasion rating into armour for the hits that do land.

This distinction carries a practical cost that gives the build its informal label: **the coward's build**. Fully realising the Invoker's defensive ceiling requires a pre-combat ritual — the *Meditate* skill, which overflows the Energy Shield pool to 1.5× its base value. A player who cannot or does not meditate before engaging a threatening encounter forfeits the primary survival buffer against large hits. This playstyle constraint is non-negotiable at high investment levels: the build is designed around strategic engagement, not facetanking.

In exchange for this constraint, the Invoker achieves gear and passive point efficiency that rivals Shaman's, through a completely different mechanism: near-zero need for life affixes, chaos resistance affixes, or spirit affixes on gear, freeing the entire affix budget for evasion and Energy Shield. The resulting eHP pool, amplified by Meditate, is conditionally larger than what most other archetypes can reach at comparable investment.

The Three Mandatory Ascendancy Nodes

The Invoker has four ascendancy choice slots. Three are near-mandatory for the defensive architecture described in this section; the fourth is available for damage or situational utility.

Meditate: The Conditional eHP Ceiling

Meditate, granted by *Faith is a Choice*, allows the Invoker to overflow their Energy Shield to 1.5× its maximum value. This overflow cannot be achieved through regener-

Table 19: **Invoker Ascendancy Nodes — Defensive Core**

Node	**Effect**	**Priority**
Faith is a Choice	Grants the Meditate skill. Doubles ES regeneration rate. ES can overflow to 1.5× maximum via Meditate between encounters.	Mandatory
Lead me through grace…	Gain spirit for each X points of combined evasion and Energy Shield on equipment. Cannot benefit from explicit spirit modifiers on equipment.	Mandatory (gate)
…and protect me from harm	35% less evasion rating (final multiplier, not overcome by increased evasion). Armour DR is calculated from combined armour *and* evasion pool.	Mandatory (key node)
Fourth Choice	Damage, utility, or situational defensive node. Outside mandatory defensive core.	Situational

"Lead me through grace… " is the gating node that unlocks "… and protect me from harm." Both must be taken to access the armour-evasion unification, making them functionally a paired investment. The "35% less" modifier on evasion is a final multiplier in the evasion rating formula and cannot be negated by increased evasion rating modifiers on passive tree or gear.

ation or flask mechanics alone — it requires actively channelling Meditate outside of combat.

The practical impact is that the Invoker operates at two distinct eHP levels:

- **Base (in combat):** The natural ES pool. This is the floor and is the value that matters if the player engages without preparation.
- **Meditate (pre-combat):** 1.5× the base ES pool. This is the ceiling and is what the build's survival thresholds are calibrated against.

Against a hit that would take 4,850 damage when it lands (derived from the armour-evasion DR layer), a character needs a base ES of 4,850 to survive without Meditate, but only 3,234 ES with Meditate active (4,850 / 1.5 = 3,233). Meditate effectively reduces the required raw ES investment by 33% for any given survival threshold — a significant return on the single ascendancy point.

The build-design implication is that the ES pool should be calibrated to the Meditate value, not the base value. Investment decisions should ask: "at 1.5× my projected ES, do I survive the relevant hit ceiling?" rather than targeting the base pool.

Chaos Inoculation and Gear Architecture

Like the Infernalist, the Invoker is the natural beneficiary of the **Chaos Inoculation** keystone passive:

$$\text{Maximum life} = 1 \qquad \text{Immune to chaos damage}$$

For the Invoker, CI delivers three compounding advantages:

1. **Chaos damage immunity.** As with the Infernalist's Altered Flesh interaction, chaos damage is the most lethal unmitigated source in current endgame content. CI converts the Invoker's greatest potential vulnerability into its most completely nullified threat.
2. **No chaos resistance required.** Capping chaos resistance conventionally consumes multiple gear suffixes and rune slots. CI frees every one of these for other affixes — at zero ongoing opportunity cost.
3. **No life affixes required.** With life set to 1 by CI, life investment is irrelevant. Every gear prefix that would otherwise go to life or life% becomes available for flat Energy Shield or increased Energy Shield.

The net effect on gear architecture: every affix slot on every piece of equipment is available for flat evasion, local evasion rating%, flat Energy Shield, and increased Energy Shield. No life, no chaos resistance, no spirit (blocked by the ascendancy). This is one of the most focused and unconstrained affix budgets of any archetype in the game.

"Lead me through grace. . . " — Spirit via Evasion and ES

This ascendancy node delivers spirit — the resource that enables active skills, auras, and minions — based on evasion and Energy Shield values on equipment, rather than from explicit spirit affixes. Simultaneously, it *prevents* the character from benefiting from explicit spirit modifiers on equipment.

The apparent detriment (no explicit spirit gear) is in practice a benefit. Spirit is a highly sought-after and contested affix on gear; other archetypes compete for it and must accept the opportunity cost of taking it over other affixes. The Invoker is *incentivised not to pursue it*, because the ascendancy generates spirit from what the Invoker already wants: high evasion and high Energy Shield values on gear. This creates a secondary reward loop:

- Pursuing evasion+ES gear (for defensive value) also passively generates the spirit budget
- That spirit budget enables *Galeforce* and other auras without consuming any affix slots
- Evasion+ES gear is generally easier to craft and less contested in trade than spirit-explicit gear
- The freed spirit affix slot becomes another evasion or ES affix

The Invoker resolves two resource pressures simultaneously (spirit and defensive stats) with a single gear pursuit. Other archetypes must solve them separately.

". . . and protect me from harm" — The Armour-Evasion Unification

This is the key defensive node and the primary reason the Invoker belongs in a document about defensive efficiency. Its two effects:

1. **35% less evasion rating** (final multiplier on the evasion formula)

2. **Armour DR is calculated from combined armour and evasion pool**

The first effect is the cost; the second is the payoff. Critically, the "less" modifier reduces evasion rating (not evasion chance directly), and the armour calculation uses the same post-less evasion value:

$$\text{Effective Evasion} = \text{Base Evasion} \times 0.65 \qquad \text{(for both dodge chance and armour)}$$

$$\text{Effective Armour} = \text{Base Armour} + \text{Base Evasion} \times 0.65$$

At 50,000 base evasion with 5,000 base armour from shields and gear:

$$\text{Effective Evasion} = 32{,}500 \qquad \text{Effective Armour} = 5{,}000 + 32{,}500 = 37{,}500$$

The character loses 35% of their evasion rating (an avoidance reduction), but gains 32,500 effective armour from the same pool. The net exchange is: *slightly fewer hits avoided, but hits that do land face meaningful DR from free armour*. At high base evasion values, the armour gain dominates the evasion loss in terms of survivability value.

DR from Evasion-Derived Armour: Hit-Size Table

The following table uses the armour DR formula ($A/(A+10H)$) applied to the effective armour pool derived from evasion, at varying base evasion investment levels. Base armour of 5,000 is assumed from shields and incidental gear.

Optimal Evasion Targets per ES Pool

The conventional armour optimisation heuristic states that effective armour should fall in the range of 12×–20× the effective HP pool to provide efficient DR against the range of hits a character encounters. For the Invoker, "effective armour" is derived from evasion, allowing a direct calculation of the evasion target per ES pool size.

The design target for an Invoker can therefore be expressed as a simple paired constraint: acquire evasion equal to 12× your base ES (at minimum) so that each point of evasion investment simultaneously improves dodge chance AND produces armour in the efficient DR range. Any evasion beyond the 20× ceiling still improves avoidance but no longer improves DR meaningfully — the investment shifts category

Table 20: **Armour DR from Evasion-as-Armour by Hit Size (5,000 base armour + evasion×0.65)**

Base Evs	Eff. Arm	H=100	H=500	H=1k	H=2k	H=4k	H=8k	H=10k
20,000	18,000	94.7%	78.3%	64.3%	47.4%	31.0%	18.4%	15.3%
30,000	24,500	96.1%	83.1%	71.0%	55.1%	38.0%	23.4%	19.7%
40,000	31,000	96.9%	86.1%	75.6%	60.8%	43.7%	27.9%	23.7%
50,000	37,500	97.4%	88.2%	78.9%	65.2%	48.4%	31.9%	27.3%
70,000	50,500	98.1%	91.0%	83.5%	71.6%	55.8%	38.7%	33.6%
100,000	70,000	98.6%	93.3%	87.5%	77.8%	63.6%	46.7%	41.2%

Cell shading uses the same tier scale as Section 3: dark green ≥95%, light green 90–94%, yellow 80–89%, orange 70–79%, red <70%. Unlike Shaman and Infernalist, whose mitigation columns hold in the yellow-green band at large hits, the Invoker's DR column falls into red above H=4,000 at all practical evasion values. This is by design: the Invoker's defence against large hits is avoidance (not being hit at all), not passive DR. The DR layer handles the inevitable hits that slip through evasion and block.

Table 21: **Optimal Base Evasion Targets by ES Pool (12×–20× DR Optimality Range)**

ES Pool	Meditate ES	Arm. Low (12×)	Arm. High (20×)	Base Evs (Low)	Base Ev
2,000	3,000	24,000	40,000	29,231	
3,000	4,500	36,000	60,000	47,692	
4,000	6,000	48,000	80,000	66,154	
5,000	7,500	60,000	100,000	84,615	
6,000	9,000	72,000	120,000	103,077	

"Base Evs" is the pre-35%-less evasion rating required so that evasion×0.65 (plus 5,000 base armour) equals the armour target. The optimality range applies to the base ES pool, not the Meditate value; Meditate is shown separately as the survivable ceiling. A 3,000 ES Invoker needs approximately 48,000–85,000 base evasion to keep armour-from-evasion in the efficient DR range. Above these evasion values, additional evasion contributes primarily to avoidance chance rather than marginal DR improvement. Meditate ES is shown to illustrate the conditional survival ceiling: a 3,000 ES Invoker with Meditate has 4,500 effective eHP at encounter entry.

rather than becoming wasted.

Survival Thresholds: When a Hit Lands

When a hit penetrates both evasion and block, the Invoker faces the raw damage reduced only by evasion-derived armour. At 50,000 base evasion (37,500 effective armour), the minimum ES required to survive each endgame hit tier is:

Table 22: **ES Survival Requirements When a Hit Lands — 50,000 Base Evasion (37,500 Eff. Armour)**

Hit Size	DR	Damage Taken	Min ES (Base)	Min ES + Meditate	Meditate Saves
1,000	78.9%	211	211	141	70
2,000	65.2%	696	696	464	232
4,000	48.4%	2,065	2,065	1,377	688
8,000	31.9%	5,447	5,447	3,631	1,816
10,000	27.3%	7,273	7,273	**4,849**	**2,424**

"Min ES + Meditate" is the minimum base ES required when Meditate overflow (1.5×) is active at encounter entry: damage taken /1.5. "Meditate Saves" is the reduction in required base ES attributable solely to the overflow. At H=10,000, Meditate reduces the required base ES from 7,273 to 4,849 — saving 2,424 ES of investment. Without Meditate, surviving the endgame hit ceiling at this evasion level would require a 7,273 ES pool, which is unrealistic for most builds. With Meditate, 5,000 ES (7,500 meditate) provides a buffer. This is the mechanical justification for the "coward's build" label: Meditate is not optional for surviving the hardest content.

The table makes explicit what the playstyle label implies: the Invoker's survival against ceiling-tier hits is *contingent* on Meditate. A 5,000 ES Invoker who enters a boss room without meditating faces a 7,273 damage hit on a 5,000 ES pool and dies. The same character who meditates has 7,500 effective eHP and survives by 227 points. The margin is not large, which is why the 12×–20× evasion optimality range and ongoing ES investment remain important — they push the meditate ceiling higher and provide more room for error.

Combined Avoidance: Evasion and Block

The Invoker's primary defence against all hits — including the largest — is not mitigating them but preventing them from connecting. With a spear and buckler setup providing near-cap block (50–65%) layered over evasion chance (50–65%), the combined hit avoidance is:

$$P(\text{hit avoidance}) = 1 - (1 - e)(1 - b)$$

Table 23: **Combined Hit Avoidance (%) — Evasion Chance × Block Chance**

Evs \ Block	30%	40%	50%	55%	60%	65%
40%	58.0%	64.0%	70.0%	73.0%	76.0%	79.0%
50%	65.0%	70.0%	75.0%	77.5%	80.0%	82.5%
60%	72.0%	76.0%	80.0%	82.0%	84.0%	86.0%
65%	75.5%	79.0%	82.5%	84.2%	86.0%	87.8%
70%	79.0%	82.0%	85.0%	86.5%	88.0%	89.5%
75%	82.5%	85.0%	87.5%	88.8%	90.0%	91.3%

At the achievable baseline of 50% evasion and 50% block, 75% of all hits are avoided entirely before any mitigation layer applies. At 65% evasion and 65% block (requiring significant investment), avoidance reaches 87.8%. Pushing both values to 75% achieves 90%+ avoidance. The 98–99% avoidance threshold implied by optimal values requires extreme investment in both stats simultaneously and may not be achievable within the passive and gear budget while maintaining the ES pool targets described above. The practical target for a well-invested Invoker is 82–88% combined avoidance, meaning 1 in 6–5 hits connects. Those connecting hits are handled by the evasion-derived armour and Meditate ES pool.

Per-Encounter Damage Expectation vs. Shaman

The Invoker and Shaman arrive at comparable per-encounter defensive outcomes through opposite mechanics. Consider 20 hits at H=1,000 in a typical encounter:

- **Shaman:** Every hit connects. Each hit takes 40.6 damage. Total encounter damage: $20 \times 40.6 = 812$.
- **Invoker (50% evs, 50% block):** 75% of hits are avoided. Expected hits connecting: $20 \times 0.25 = 5$. Each landing hit takes 211 damage. Total encounter damage: $5 \times 211 = 1{,}055$.

On expected value, the Invoker takes slightly *more* total damage per encounter at H=1,000 (1,055 vs 812). However, the Invoker's defence profile produces dramatically different variance: most encounters end with zero damage (all hits avoided), but the build is exposed to burst clustering where multiple consecutive hits land before evasion resets. Shaman's profile is flat — it takes the same amount of damage regardless of hit sequencing.

This is the core trade-off: Shaman is statistically smooth; Invoker is statistically bursty.

The Invoker mitigates burstiness through Meditate's pre-combat eHP overhead and block's independent avoidance layer, but the risk profile is fundamentally different. At H=10,000 (the ceiling where survival matters most), the Invoker with full Meditate is not guaranteed to survive a landed hit unless the ES pool and evasion-armour are correctly calibrated.

Supporting Systems and Future Exploration

Several additional mechanics compound the Invoker's defensive and offensive efficiency but fall outside the quantitative scope of this document:

- **Galeforce:** An aura that grants stacking evasion rating increases, funded by the spirit pool generated passively through "Lead me through grace…" Because the spirit pool scales with gear evasion+ES values (which the Invoker is already maximising), Galeforce is effectively a zero-cost aura. Its stacks push base evasion rating higher during combat, improving avoidance and increasing the effective armour value in the armour-evasion formula.
- **Deflection:** A mechanic that applies a front-end multiplicative reduction to hit damage before armour is calculated. If accessible to the Invoker, deflection would reduce the effective hit size entering the armour formula, improving DR at every hit tier — analogous to how physical-to-elemental conversion reduces the effective hit size for Shaman. Exploring the interplay between deflection and evasion-derived armour is a priority for future analysis.
- **Hollow Palm:** A keystone or technique available to unarmed Monk builds that scales defences (and damage) from having no weapon equipped. Whether a Hollow Palm Invoker can achieve comparable or superior evasion-armour targets through a different gearing axis is worth investigating, particularly given the freed gear slots.
- **Spear and Buckler Synergy:** The reference configuration (spear + buckler) provides the near-50% block baseline. Combined with the evasion values described above, this produces the 82–88% avoidance band. Exploring whether shield-focused passive nodes can push block toward 60–65% while preserving evasion investment is likely the highest-leverage path to improving the Invoker's ceiling, as 65% evasion + 65% block (87.8% avoidance) represents a substantially different risk profile than the 50%+50% baseline.

Invoker Positioning

Table 24: **Invoker vs. Shaman: Defensive Profile Comparison**

Property	**Shaman**	**Invoker**
Primary defence	Passive hit mitigation (conversion + layers)	Avoidance (evasion +
DR on a landed hit (H=10k)	80.8%	27.3% (50k evs) – 41.2
eHP source	Life (+ MOM/EB optional)	Energy Shield (CI mar
eHP conditional boost	Mind over Matter (passive)	Meditate (active pre-c
Chaos damage	Near-zero via resistance	Zero (CI immunity)
Gear affix budget	Frees armour affixes	Frees life, chaos res, sp
Passive point spend	Damage nodes (conversion from ascendancy)	Evasion+ES nodes (de
Playstyle constraint	None	Meditate before challe
Ceiling comparison	Survives H=10k on 1,923 HP (passive)	Survives H=10k with
Verdict	**Passive survivability, lower eHP floor**	**Conditional higher**

The two archetypes are not directly comparable by a single metric. Shaman guarantees survival of the H=10,000 ceiling on 1,923 HP passively. Invoker guarantees avoidance of ∼75–88% of all hits, with survival of landed hits conditional on Meditate. The Invoker's strength is encounter-level expected damage (most hits miss) and a larger achievable eHP ceiling via Meditate overflow; Shaman's strength is unconditional per-hit survivability and passive simplicity. Players optimising for smoothness of play and reliability under all conditions favour Shaman; players optimising for peak eHP ceiling and total encounter damage intake under optimal play favour Invoker.

8 Warbringer: Block-as-Mitigation and the Mapping Specialist

Class Overview and Build Identity

The Warbringer is an ascendancy of the Marauder base class and represents an interesting adjacent case to Smith of Kitava: both are Marauder-lineage archetypes that lean into armour, block, and physical survivability, but the Warbringer resolves the armour formula's scaling weakness through a different structural intervention. Rather than using conversion or absorption, the Warbringer converts block chance into a front-loaded damage multiplier — reducing hits that connect through its block roll to 20% of their original value before armour evaluates. Combined with a totem-linked damage reduction layer, the Warbringer achieves excellent *expected* mitigation across the hit-size spectrum. Its constraint is variance: the 25% of hits that bypass block land at 80% of their value and expose the same fundamental armour formula weakness that limits every non-converting archetype at large hit sizes.

The build's natural home is totem-based content: high-uptime totems enable the secondary mitigation layer (Wooden Wall) near-continuously, and the ranged playstyle reduces the frequency of hits that reach the character at all. Against boss-tier content with single large strikes, the probabilistic nature of block creates the defining tension — one the section explores in detail.

The Two Core Defensive Nodes

Combined Multipliers: The Blocked-Hit Arithmetic

With both nodes active and a hit successfully blocked, two front-loaded multipliers apply in sequence:

$$\text{Turtle Charm (blocked): } \times 0.20 \qquad \text{Wooden Wall (totem up): } \times 0.80$$

$$\text{Pre-armour effective hit (blocked): } = 0.20 \times 0.80 \times H = 0.16H \quad \Rightarrow \quad 84\% \text{ less damage before armour}$$

On an unblocked hit with Wooden Wall active:

$$\text{Pre-armour effective hit (unblocked): } = 0.80H \quad \Rightarrow \quad 20\% \text{ less damage before armour}$$

Table 25: **Warbringer Ascendancy Nodes — Defensive Core**

Node	Effect	Priority
Turtle Charm	Maximum block raised to 75%. You take 20% of the damage of blocked hits (80% less — confirmed front-loaded). Enables blocking of previously unblockable hits.	Mandatory
Wooden Wall	20% of damage you would take is instead taken by your totem. Requires an active totem.	Mandatory
Third/Fourth Choice	Offensive scaling, totem enhancement, or utility investment.	Situational

Turtle Charm's 80% less on blocked hits is confirmed to apply before the armour formula — the armour formula evaluates against 20% of the original hit, not the full hit. Wooden Wall's 20% reduction applies simultaneously, reducing the pre-armour value further. Both multipliers apply regardless of hit type (physical, elemental, chaos) provided the blocking condition and totem condition are met. Near-100% totem uptime is assumed throughout this analysis given the Warbringer's identity as a totem build.

The implications for the armour formula are stark. Armour is highly efficient against small hits (the formula's numerator dominates when $H_{\text{eff}} \ll A$) and poor against large hits (the denominator grows with H). By collapsing a 10,000 hit to a 1,600 effective hit on a blocked roll, the Warbringer places armour in its most efficient operating regime 75% of the time — while the other 25% exposes armour to an 8,000 effective hit it cannot handle as efficiently.

Hit-Size Survival Table

The following table uses A = 50,000 armour with near-100% totem uptime (Wooden Wall always active). $P(\text{block}) = 0.75$.

The expected value result — 84.8% mitigation surpassing Shaman's deterministic 80.8% — is the build's genuine appeal in mapping content where many hits arrive and the block distribution averages out. Against a single boss slam where one non-block is fatal, the expected value is irrelevant; the 25% non-block probability is the only number that matters.

Table 26: **Warbringer Damage Outcomes at A=50,000 (Turtle Charm + Wooden Wall Active)**

Hit	Eff. Hit (Blk)	Blk Dmg	Eff. Hit (No Blk)	No Blk Dmg	Expected	Eff. Mit.
500	80	1	400	30	8	98.3%
1,000	160	5	800	110	31	96.9%
2,000	320	19	1,600	388	111	94.4%
4,000	640	73	3,200	1,249	367	90.8%
8,000	1,280	261	6,400	3,593	1,094	86.3%
10,000	1,600	388	8,000	4,923	1,522	84.8%

"Eff. Hit" is the effective hit size entering the armour formula after front-loaded multipliers. "Blk Dmg" and "No Blk Dmg" are the final damage values on each outcome. Expected $= 0.75 \times$ *Blk Dmg* $+ 0.25 \times$ *No Blk Dmg. At H=10,000, the expected mitigation of 84.8% exceeds Shaman's passive 80.8% — a counterintuitive result driven by armour's high efficiency against the 1,600 effective blocked hit. The 4,923 unblocked damage on the final row is the class-defining constraint: surviving this outcome requires 4,924 effective HP, more than double Shaman's 1,924 HP requirement.*

Armour Targets: The eHP Constraint

Unlike other conversion or absorption archetypes in this paper where armour recedes as a binding constraint, the Warbringer's planning target for worst-case survival is dictated jointly by armour and eHP. The armour required to guarantee survival of an unblocked $H = 10{,}000$ hit falls steeply as the life pool grows:

$$A^* = \frac{10 \times 0.8H \times (0.8H - L)}{L}$$

The Unblockable Hit Interaction

Turtle Charm enables the Warbringer to block hits flagged as unblockable — including boss slams and telegraphed area attacks that every other archetype must dodge manually or absorb at full value. This is the build's most uniquely powerful defensive property and has no equivalent in the other archetypes examined in this paper.

The practical evaluation is probabilistic. At 75% block on a boss slam of magnitude H:

- **75% of occurrences:** Hit is blocked. Pre-armour damage $= 0.16H$. Armour handles a tiny effective hit efficiently. The slam is survived trivially.
- **25% of occurrences:** Hit is not blocked. Pre-armour damage $= 0.80H$ (Wooden

Table 27: **Armour Required to Survive Unblocked H=10,000 (Wooden Wall Active, by Life Pool)**

Life Pool (L)	Eff. Hit (0.8H)	Required Armour	Multiple of L
3,000	8,000	133,333	44.4×
4,000	8,000	80,000	20.0×
5,000	8,000	48,000	9.6×
6,000	8,000	26,667	4.4×
8,000	**8,000**	**0**	**0**

The armour requirement falls quadratically with life pool growth. At $L = 3{,}000$, the armour required (133k) is not achievable within normal gearing. At $L = 5{,}000$, 48k armour — achievable on a dedicated armour build — guarantees survival. At $L = 8{,}000$ (equal to the effective unblocked hit), zero armour is needed: the pool absorbs the hit regardless. This table is the formal justification for the user's observation that the Warbringer's core constraint is eHP, not armour: raising the pool from 3k to 5k reduces the armour requirement from impractical (133k) to standard (48k). The recommended build path is an armour/Energy Shield hybrid where ES fills the pool to the 5k–6k range, enabling armour targets that are achievable through normal investment.

Wall only). This is a large effective hit and survival depends entirely on whether the life pool and armour together absorb it.

Against a 10,000 raw slam at the gear levels in the table above: the 25% non-block outcome deals 4,923 damage at A=50k or 3,556 damage at A=100k. Whether these values are survivable is entirely pool-dependent. On a 5,000 HP pool: a 4,923-damage outcome is survived by 77 HP at A=50k; a 3,556-damage outcome is survived comfortably at A=100k. On a 3,000 HP pool: neither is survivable.

The user's framing is precise: "you are playing an RNG game where you hope that you block the slam." The build offers a 3-in-4 success rate on the worst hits in the game — a genuine advantage over builds that must dodge manually and rely on player execution — but cannot guarantee survival without a sufficient eHP pool behind the 25% failure case.

Build Path: Standard Block vs. Turtle Charm

A Warbringer who skips Turtle Charm retains standard block mechanics (50% chance, full immunity on blocked hits) and saves two ascendancy points. With Wooden Wall and a terminal 10% less multiplier from another source:

- **P(block) = 0.50:** Full immunity — 0 damage taken.

- **P(no block) = 0.50:** $0.80H$ from Wooden Wall, then armour DR, then $\times 0.90$ terminal multiplier.
- **Expected (A=50k, H=10k):** $0.50 \times 0 + 0.50 \times 4{,}430 = 2{,}215$ — 77.9% expected mitigation.

Against Turtle Charm's 1,522 expected damage (84.8% mitigation), the standard block path takes 46% more expected damage per hit at equivalent armour. The unblocked worst-case damage is identical (the armour formula sees the same 0.8H), but the probability of triggering it is 50% rather than 25%. The trade-off for the two saved ascendancy points — which can go to offensive or totem-enhancing nodes — is a substantially higher expected damage intake per encounter and twice the probability of the worst-case outcome. Turtle Charm is the correct defensive choice when the ascendancy budget allows it.

Manual Block and the Active Defense Variant

The probability table in the preceding section rests on one assumption: block is passive and probabilistic at 75%. A single condition inverts this entirely. Path of Exile 2 provides active block skills — specifically **Raise Shield** and **Resonating Shield** — that allow a player to guarantee a block on demand. When an active block is executed correctly, the Warbringer's block chance goes from 75% (passive RNG) to 100% (player-controlled), and Turtle Charm's 80% less multiplier becomes guaranteed rather than probabilistic.

In mapping, this changes nothing in practice — the passive 75% rate is sufficient and manual blocking every pack hit would be exhausting. In boss encounters, it transforms the build: instead of hoping that a boss slam rolls into the 75% window, the player manually blocks the telegraphed strike and converts a potential one-shot into trivial damage ($0.16H$ before armour). Failure to execute the block still results in taking the full unmitigated hit, making it a skill-check rather than an RNG outcome.

Two mutually exclusive ascendancy paths. Turtle Charm is mandatory on either path. The second ascendancy node allocates to either Wooden Wall (the totem route) or Jade Heritage — these are not additive, and the build takes one or the other.

Wooden Wall path: the totem as an independent DPS engine. The strategic advantage of Wooden Wall extends beyond the 20% redirect. The Warbringer's combat totems deal damage independently and automatically, requiring no player attention once placed. This creates a structural separation between the build's damage source

Table 28: **Wooden Wall vs. Jade Heritage: Constant Damage Reduction Comparison**

Property	Wooden Wall (Totem Route)	Jade Heritage
Constant reduction	20% of damage → totem; player takes 80%	10% less terminal; player takes 90%
Conditional mechanic	Requires active totem (near-100% uptime on totem build)	Guard skill: 60% of life as absorptive shield, on cooldown
Totem dependency	Yes — uptime is required for the layer to function	No
Verdict	**Strictly superior constant reduction**	**Guard provides conditional buffer; weaker continuous reduction**

The constant-reduction comparison is unambiguous: Wooden Wall routes 20% of every hit to the totem while Jade Heritage reduces only 10% terminally. Jade Heritage's Guard skill provides a conditional absorptive buffer equal to 60% of maximum life on a cooldown — but this is a reactive tool with a recharge period, not continuous damage reduction. The Encased in Jade interaction is resolved: the Encased in Jade state and manual block are mechanically exclusive — a player cannot raise their shield while encased in jade. Passive RNG block can still proc while encased (applying Turtle Charm's 80% less on those rolls), but this is unreliable and does not restore the deterministic control that defines the manual block ceiling. The 10% vs. 20% differential holds in all practical scenarios.

and the player character that closely mirrors how minion builds operate: the totem runs its damage loop while the player allocates all attention to defensive execution — manual blocks on telegraphed strikes, dodged mechanics, and positioning. The opportunity cost of taking Wooden Wall over Jade Heritage is therefore near-zero from a damage perspective: the totem enabling the mitigation layer is the same totem dealing damage.

This architecture extends naturally to a weapon-swap variant. In one weapon set: combat totems as the primary damage source, enabling Wooden Wall's mitigation layer continuously. In the second weapon set: a shield for active blocking, optionally combined with a minion-summoning weapon. The result is two independent damage engines — totems and minions — running simultaneously while the player focuses entirely on defensive execution. Both damage sources require no player input once deployed. The build effectively plays itself offensively while the player plays it defensively — the same division of responsibility that defines effective minion builds at the skill ceiling.

Jade Heritage as the alternative. Jade Heritage is suboptimal for one-shot protection but situationally valid for specific content types. The decisive mechanical constraint: **Encased in Jade and manual block are exclusive** — a player cannot raise their shield while encased in jade. Taking Jade Heritage therefore means abandoning the manual block ceiling entirely. Passive RNG block can still proc while encased (applying Turtle Charm's 80% less on those rolls), but it is unreliable and does not restore the deterministic control that defines the Wooden Wall path's boss-kill value.

Jade Heritage's Guard skill is most effective in sustained medium-hit content: many hits of moderate size that collectively threaten the life pool rather than a single catastrophic slam. Guard activates a temporary absorptive shield equal to 60% of maximum life (= 2,700 at $L = 4{,}500$) ahead of the life pool on a cooldown that refills between waves of hits, providing a repeatable buffer against cumulative damage intake. Against a single large hit, Guard is a one-time buffer that may not cover the full damage and cannot refill before the next slam.

The stun risk of manual blocking is also a legitimate factor: committing to a block action leaves the player exposed if the block is mis-timed or fails, and a stun in that window creates a vulnerability to immediate follow-up hits. The Encased in Jade state provides stun protection that has real defensive value in high-density encounters where a stun from a medium hit followed by a large hit is the actual kill condition. Against that threat profile specifically — sustained pressure with stun-chain risk, not single-hit one-shot ceiling — Jade Heritage is the correct choice. Its argument is not improving one-shot protection; it is managing the sustained-damage environment where stun is the primary threat.

Veil of Night and the HP breakpoint. Achieving the eHP thresholds identified in the armour target table ($L \geq 5{,}000$ for 48k armour; $L \geq 4{,}000$ for 80k armour) is a gear-budget problem. *Veil of Night* resolves this directly: it sets all elemental resistances to 0% and grants 50% increased maximum life. On a 3,000 base life pool, Veil of Night forces $L = 4{,}500$ — a gain of 1,500 HP that reduces the worst-case armour requirement from 133k to approximately 68k. The lost elemental resistances must be recaptured through other slots, but the net trade is often favourable: 1,500 HP contributes more survivability than the 350–1,000 flat armour a conventional rare chest would provide, and it frees the chest's suffixes for resistance or offensive affixes. *Hollow Mask* is the alternative: it preserves resistances while providing additional physical damage reduction, suited to players who cannot cap resistances elsewhere but want to stay in the armour-building path.

Updated verdict under active play. The Warbringer with Turtle Charm, Wooden Wall, and manual blocking on boss-tier hits is the recommended high-skill-expression configuration. In mapping, passive 75% block with Wooden Wall already produces better expected mitigation than Shaman (84.8% vs. 80.8% at A=50k); with manual block available on telegraphed slams, those hits leave the probability distribution entirely and become deterministic low-damage events. The totem provides continuous DPS with no player input required, and a weapon-swap to a shield set with minion support layers a second independent damage engine behind the same defensive stance. The build's ceiling is constrained by player execution and achievable eHP, not by armour formula limits or absorption pool caps. This raises the skill floor above any other archetype examined in this paper — but it rewards that execution with the highest expected mitigation of any armour-anchored build and deterministic worst-case control over telegraphed boss-tier hits.

Warbringer Positioning

Table 29: **Warbringer vs. Smith vs. Shaman at H=10,000**

Property	Shaman	Smith of Kitava	Warbringer (A=100k)
Mitigation mechanism	Conversion	Pure armour	Block + front-loaded mu
Expected damage	1,923 (deterministic)	5,000	1,055
Worst-case damage	1,923	5,000	3,556
Min HP (worst case)	1,924	5,001	3,557
Armour investment	Minimal (conversion floor)	100k+	100k+
Chaos coverage	Near-zero residual	Dedication to Kitava	Armour DR only
Blocks unblockable hits	No	No	Yes (Turtle Charm)
Totem dependency	No	No	Near-mandatory (Woode
Best content	General endgame	Crit/chaos maps	Mapping, totem content
Verdict	**Lowest HP floor, passive**	**Crit/chaos specialist**	**Best expected value;**

Warbringer at A=100k achieves the best expected mitigation of the three armour-anchored builds (89.5% vs Shaman's 80.8% vs Smith's 50%). This result is counterintuitive but mathematically sound: armour is extraordinarily efficient against the 1,600 effective blocked hit (86.2% DR), dragging the expected damage well below Shaman's passive threshold. The cost is the 3,557 HP minimum for the worst-case unblocked hit — a requirement that Shaman never faces because its passive mitigation makes the worst case 1,923 regardless of block outcomes. The Warbringer is genuinely the best Marauder-lineage defensive archetype examined in this paper, but it does not match the investment efficiency of Shaman, Infernalist, or Witchhunter, which achieve superior or comparable expected mitigation at a fraction of the armour investment and without the life pool constraint imposed by the 25% unblocked outcome.

9 Tactician: Deflection, Capped DR, and the Armour-Efficient Paradigm

Class Overview and Build Identity

The Tactician is the second ascendancy of the Mercenary base class and occupies an unusual position in the defensive landscape: its core mechanical identity — supporting totems, crossbows, and minions at range — is intelligible from the passive tree, but the defensive nodes required to access that identity consume the first two ascendancy slots before any offensive expression is possible. For most players, the Tactician is a build that becomes itself only after the campaign ends. This deferred payoff is not a mechanical flaw; it is a design constraint worth naming, because it shapes how one should evaluate the class's defensive architecture independent of its playstyle.

The two near-mandatory defensive nodes are **Polish that Gear** and **Stay Light, Use Cover**. Together they define an archetype with a hard DR ceiling, a front-loaded damage reduction mechanic, and strong pressure toward ranged engagement. They also create a genuine tension in build optimisation: the DR cap imposed by Stay Light limits how much armour investment matters, the evasion cap it imposes is compounded by the Glancing Blows keystone, and the primary defensive win condition — deflection — depends on a probabilistic chain that the build can influence but never fully control.

The Two Core Defensive Nodes

Deflection: The Front-Loaded Multiplier

Polish that Gear grants deflection rating equal to 20% of armour rating. When a hit is deflected, damage is reduced by 40%. The operative question for this analysis — one that the game's tooltip does not resolve — is whether this reduction applies *before* or *after* the armour formula evaluates.

- **If front-loaded (deflect reduces hit size entering armour):** The armour formula evaluates against $0.6H$ rather than H. The armour formula produces better DR on a smaller effective hit, and the ordering strictly improves outcomes when armour is below the DR cap.
- **If back-loaded (deflect reduces the post-armour remainder):** The armour formula evaluates against the full H. Deflect applies as a multiplier on the damage taken after armour DR.

Table 30: **Tactician Ascendancy Nodes — Defensive Core**

Node	Effect	Priority
Polish that Gear	Deflection rating = 20% of armour rating. Ailment threshold = 100% of evasion rating.	Mandatory
Stay Light, Use Cover	Defend with 200% of armour. Enemies have an accuracy penalty based on distance. Maximum chance to evade = 50%. Maximum damage reduction = 50%.	Near-mandatory
Third/Fourth Choice	Build-defining offensive or totem/minion nodes — accessible only after the defensive core is purchased.	Situational

"Polish that Gear" is a prerequisite for "Stay Light, Use Cover." The "Near-mandatory" label on Stay Light reflects the author's assessment: the 200% armour defend modifier halves the armour investment required to reach any given DR, and the accuracy-from-distance penalty supports the crossbow/totem identity. A viable alternative skipping Stay Light is analysed in the Build Path Divergence section.

At the 50% DR cap imposed by Stay Light, the two orderings produce **identical results**:

$$\text{Front-loaded: } 0.6H\times(1-0.50) = 0.30H \qquad \text{Back-loaded: } H\times(1-0.50)\times 0.60 = 0.30H$$

The analysis below treats the DR cap as the planning target, where ordering is irrelevant. Below the cap, front-loading is strictly better: the armour formula operates on a smaller effective hit and yields higher DR at every armour value. A build that consistently reaches the 50% DR cap on relevant hits loses nothing to the ambiguity; one that does not reach the cap benefits more if deflect is front-loaded.

Effective DR on a deflected hit at the cap:

$$\text{Damage taken} = 0.6H \times 0.50 = 0.30H \quad \Rightarrow \quad \text{Effective DR} = 70\%$$

Effective DR on an undeflected hit at the cap:

$$\text{Damage taken} = H \times 0.50 = 0.50H \quad \Rightarrow \quad \text{Effective DR} = 50\%$$

Armour Targets Under 200% Defend

The 200% armour defend modifier means the DR formula evaluates 2A against the incoming hit, not A. The standard corridor (12–20×L) was derived for the unmodified formula; it does not apply here.

$$DR = \min\left(0.50,\ \frac{2A}{2A + 10H}\right)$$

Cap condition (no deflect): $2A \geq 10H \Rightarrow A \geq 5H$. The armour required to guarantee 50% DR against a raw hit H is exactly $5H$ — half of the standard zero-conversion requirement of $10H$.

Cap condition (with deflect): The formula evaluates against $0.6H$, so $2A \geq 6H \Rightarrow A \geq 3H$.

Table 31: **Revised Armour Corridor — Tactician (200% Defend, 50% DR Cap)**

Scenario	Cap Condition	Vs 2L Kill Threshold	Effective Corridor
Standard (no mod)	$A \geq 10H$	$A = 20L$	12×–20 × L
Tactician, no deflect	$A \geq 5H$	$A = 10L$	5×–10 × L
Tactician, with deflect	$A \geq 3H$	$A = 6L$	3×–6 × L
Planning target (worst-case)	No deflect	$A = 10L$	≈ 8×–10 × L

"Kill threshold" assumes 50% DR cap: maximum survivable hit = $2L$ *(half is taken, half is absorbed by pool). Armour required for 50% DR against that hit:* $5 \times 2L = 10L$ *(no deflect) or* $3 \times 2L = 6L$ *(with deflect). Because deflection is probabilistic, the prudent planning target is the no-deflect corridor of 8–10×L. This is a structural reduction from the standard 12–20×L heuristic: the 200% defend modifier provides the equivalent of doubled armour investment, halving the gearing cost to reach the DR cap. Additional armour beyond the cap condition yields zero marginal DR against the planning hit, because the 50% cap is reached.*

Glancing Blows and the Sequential Probability Chain

Glancing Blows is a keystone passive that inverts the luck of two defensive stats simultaneously: evasion rolls become *unlucky* (roll twice, take the worse) and deflection rolls become *lucky* (roll twice, take the better). For the Tactician, whose evasion is hard-capped at 50% by Stay Light and whose deflection is the primary win condition, Glancing Blows is the correct keystone — but it is important to understand what the tradeoff actually costs and pays.

Unlucky evasion at 50% cap:

$$P(\text{evade}) = P(\text{both rolls succeed}) = 0.50 \times 0.50 = 0.25$$

Lucky deflection at base chance d:

$$P(\text{deflect}) = 1 - (1 - d)^2 = 2d - d^2$$

At $d = 0.50$: $P(\text{deflect}) = 0.75$. At $d = 0.60$: $P(\text{deflect}) = 0.84$.

Glancing Blows trades 25 percentage points of evasion effectiveness (50% $\rightarrow$ 25% effective) for a large improvement in deflection rate. Whether this trade is favourable depends on the relative value of each avoidance layer, which the probability chain makes explicit.

Causal chain on an incoming hit:

1. Hit connects (accuracy check passes against evasion).
2. 25% chance to evade — hit is negated entirely. $P = 0.25$.
3. If not evaded (75%), deflection roll. At $d = 0.50$: 75% chance to deflect. $P(\text{not evade, deflect}) = 0.75 \times 0.75 = 0.5625$.
4. If not deflected (25% of non-evaded): armour alone. $P(\text{not evade, not deflect}) = 0.75 \times 0.25 = 0.1875$.

The survival implication of the worst-case row is the class's defining constraint: the 18.75% of hits that land without deflection require 5,001 effective HP. This is structurally identical to Smith of Kitava's requirement, but Tactician reaches this number on a mixed armour-evasion chassis that is competing for affixes across more defensive stat categories. Shaman, by contrast, requires only 1,923 HP to guarantee survival of the same hit — passively, with no probability chain.

Table 32: **Expected Damage Taken at H=10,000 — Tactician (A=50,000, 25% Eff. Evasion, 75% Eff. Deflect)**

Outcome	Probability	DR	Damage Taken	Contribution
Evaded	25.0%	100%	0	0
Not evaded, deflected	56.25%	70.0%	3,000	1,688
Not evaded, not deflected	18.75%	50.0%	5,000	938
Expected	**100%**	**73.8%**	—	**2,625**

A=50,000 reaches the 50% DR cap on undeflected hits at H=10,000 ($2 \times 50k = 100k \geq 10 \times 10k$*). Deflected hits take 0.6×10,000 at 50% DR = 3,000. Non-deflected hits take 10,000 at 50% DR = 5,000. The worst-case (non-evaded, non-deflected) requires 5,001 effective HP to survive — identical to Smith of Kitava's requirement at the same hit ceiling, but reached on a class whose gear and passive budget is split across armour, evasion, life, and resistances. Expected mitigation of 73.8% trails Shaman's passive 80.8% despite heavier defensive investment.*

The Ailment Threshold Benefit

Polish that Gear grants ailment threshold equal to 100% of evasion rating. Ailment threshold determines how much of a hit's damage must land before an ailment is applied. At high evasion investment (50,000–70,000 evasion rating), the threshold is large enough that:

- **Shock and freeze** — which require a hit to exceed a threshold proportional to maximum life — become functionally intermittent. At sufficient evasion investment, most endgame hits fall below the threshold and these ailments apply so rarely they cease to be a defensive concern.
- **Ignite, bleed, and poison** — damage-over-time ailments not governed by threshold in the same way — still require charm-based removal.

This is not immunity; it is functional suppression of two of the most punishing ailments in endgame content (Shocked ground, freeze during a dangerous encounter) at the cost of evasion investment the build is already pursuing. The ailment suppression is a compounding secondary benefit, not the primary reason to invest in evasion.

Build Path Divergence: Evasion Hybrid vs. Block-and-Armour

The foregoing analysis assumes the standard Tactician build: evasion capped at 50% (unlucky to 25%), deflection from 20% of armour, and Stay Light, Use Cover providing 200% armour defend. A second path is viable — and in some respects more internally consistent — that forgoes evasion entirely:

Table 33: **Tactician Build Path Comparison**

Property	**Evasion Hybrid (Standard)**	**Block-and-Armour (Alt)**
Ascendancy nodes	Polish that Gear + Stay Light	Polish that Gear only
Avoidance layer	25% eff. evasion (unlucky 50%)	50% block
DR cap	50% (imposed by Stay Light)	None (uncapped armour DR)
Armour defend	200%	Standard (100%)
Armour target	$5\times$–$10 \times L$	$10\times$–$20 \times L$ (higher, no modifier)
Deflect source	20% of armour	20% of armour (same)
DR on large hits (H=10k)	50% hard cap	Uncapped; scales with armour inves
Ailment threshold	Yes (evasion-dependent)	Reduced (less evasion)
Gear budget pressure	High (armour, evasion, life, res)	Moderate (armour, life, res)
Win condition	**Probabilistic: evade or deflect**	**Heavy deflect + high armour D**

The case for the Block-and-Armour path: Forgoing Stay Light, Use Cover frees two ascendancy points, removes the 50% DR cap, and removes the evasion cap. Replacing evasion with block (50% block = 50% avoidance, not penalised by Glancing Blows) provides equivalent or superior avoidance compared to the unlucky 25% effective evasion the hybrid path achieves. With heavy armour investment, the deflection rate (lucky from Glancing Blows at 20% of armour) rises with armour — both deflect chance and raw armour DR increase simultaneously. If deflection applies front-loaded, a deflected hit at high armour values can achieve 60–70% armour DR on the $0.6H$ remainder, producing a total effective DR of 76–82% on deflected hits — approaching Shaman's passive 80.8%.

The unresolved constraint: Both paths depend on the same unknown: the actual deflection chance formula. If deflection rating of $0.2A$ at, say, $A = 100{,}000$ (deflect rating = 20,000) produces a high base chance, the lucky Glancing Blows roll yields a reliable 85%+ deflect rate, and the Block-and-Armour path is efficient. If deflection rating scales poorly or has a hard cap unrelated to armour, the premise collapses. This is an empirical question — not resolvable from the tooltip text alone — and represents the primary measurement gap in this section's analysis.

The Identity Crisis: Optimisation Without a Dominant Constraint Solver

The Tactician's core tension is this: the build imposes two independent scaling requirements (armour large enough to matter and evasion large enough for ailment threshold and avoidance) without providing a mechanism that converts one into the other at efficiency. The Invoker's "...and protect me from harm" node converts evasion directly into armour. The Witchhunter's Sorcery Ward converts armour into an absorption pool. The Tactician converts armour into deflection rating (20%) — a valuable secondary benefit, but one whose payoff depends on a probability distribution rather than a guaranteed quantity.

This produces the optimisation problem the user correctly identifies: you must plan armour for the worst-case (no deflect) and plan life pool for the worst-case (no evade, no deflect), but the build is rewarded most when both fire. The character who deflects consistently takes 3,000 from a 10,000 hit on 50,000 armour; the character who does not deflect takes 5,000 from the same hit and needs 5,001 effective HP to survive. These two planning targets do not converge.

The Witchhunter resolves this tension structurally: Sorcery Ward absorbs the post-armour remainder whether or not a deflection occurred, providing a deterministic secondary layer. The Tactician has no equivalent. Its survival of extreme hits is conditional on the probability chain resolving favourably, which is an inherent property of the archetype rather than a gap correctable through investment.

Tactician Positioning

Table 34: **Tactician vs. Shaman vs. Witchhunter at H=10,000**

Property	**Shaman**	**Witchhunter**	**Tactician**
Mitigation mechanism	Conversion (DR suppression)	SW absorption	Deflect + c
Expected damage (H=10k)	1,923 (passive)	0 (SW intact)	2,625 (prob
Worst-case damage	1,923 (deterministic)	5,000+ (SW depleted)	5,000 (no d
Min HP to survive worst case	1,924	3,001 (+ SW buffer)	5,001
Chaos coverage	Near-zero residual	SW absorbs post-DR	Armour D
Armour target	Minimal (conversion does work)	$6\times$–$10\times L$ (+ SW)	$8\times$–$10\times L$ (
DR cap	None (conversion asymptote)	50% (post-penalty) + SW	50% hard c
Unique item required	Yes (Cloak of Flame)	No	No
Build expression delay	None	None	Post-campa
Verdict	**Lowest HP floor, passive**	**Highest one-shot ceiling**	**Efficient a**

Tactician expected mitigation (73.8%) trails Shaman (80.8%) despite heavier combined investment, because the 50% DR cap limits the ceiling on undeflected hits and the mixed gear requirements constrain life pool depth. Its structural advantage is armour efficiency: the 200% defend modifier halves the armour required to reach the DR cap versus a zero-conversion build, making the armour target genuinely achievable. The build is rewarding under the probability chain — deflect + DR produces 70% effective DR, comparable to Shaman — but cannot guarantee that chain fires on every hit, which is Shaman's unconditional guarantee.

10 Smith of Kitava: Armour, Conversion, and the Critical Strike Frontier

Class Overview

The Smith of Kitava is an ascendancy of the Marauder base class and occupies a specific position in the defensive hierarchy: the armour-primary archetype that, through dedicated passive investment, partially escapes the formula's structural ceiling by extending armour's coverage to the converted elemental portion of incoming hits. Unlike the pure conversion archetypes (Shaman, Infernalist) that reduce the effective hit size entering the formula, Smith of Kitava keeps the hit whole and expands what the formula evaluates against.

Two mechanisms define the Smith's defensive model:

1. **"X% of Armour applies to Elemental Damage" stacking.** Multiple passive sources accumulate additively, making the effective armour-to-elemental modifier (Y) exceed 1.0 through dedicated investment. Paired with a 35% physical-to-elemental conversion source, both the physical and elemental components of the

split hit are evaluated against the same armour pool.

2. **Blade Catcher and the critical strike frontier.** Smith of Kitava's core ascendancy node converts incoming critical strikes to base damage (removing the critical multiplier) and defends those hits at 200% of armour rating — effectively $2A$ in the DR formula. In critical-strike-heavy map content, this doubles the armour value on the hits that would otherwise be most lethal, creating a content-specific ceiling no other archetype in this paper matches.

The Smith occupies B-Tier in this paper's ranking: its achievable VHP multiplier ($2.5\times$ at $Y = 1.0$, $A = 80{,}000$) exceeds the Armour Pure baseline ($2.0\times$) through conversion and Y-stacking, but trails the S-Tier conversion archetypes at general endgame investment. Its niche is specific and genuine: in content where critical strikes are the primary threat, no other archetype matches the Smith's structural advantage.

Optimal Smith Configuration: Armour-to-Elemental and Veil of Night

The comparative analysis in Section 3 treated Smith of Kitava as a pure-armour archetype with no elemental handling layer. A structurally superior configuration adds two elements: passive investment in "X% of Armour applies to Elemental Damage" nodes and a 35% physical-to-elemental conversion source. The armour target remains the 12–20$\times L$ corridor that binds all armour-based builds; the configuration does not change the planning constraint, it changes what each armour point accomplishes.

With 35% conversion, the incoming hit splits into a physical component of $0.65H$ and an elemental component of $0.35H$. The elemental component evaluates against armour scaled by the armour-to-elemental percentage Y:

$$\mathrm{DR}_{\mathrm{phys}} = \frac{A}{A + 6.5H} \qquad \mathrm{DR}_{\mathrm{ele}} = \frac{Y \cdot A}{Y \cdot A + 3.5H}$$

Y is not capped at 1.0 — multiple "X% of Armour applies to Elemental Damage" passive sources stack additively, making $Y = 1.5$–2.0 and higher achievable through dedicated investment. The armour required to reach 75% elemental DR (the threshold above which elemental resistances become optional) scales inversely with Y:

$$A_{75\%} = \frac{10.5 \cdot H}{Y}$$

Table 35: **Armour Required for 75% Elemental DR by Y (35% Conversion, $H = 10{,}000$ Ceiling)**

Y **(Armour-to-Elemental)**	**Armour Required**	**Min L for Corridor** ($A \leq 20 \times L$)	
1.0 (100%)	105,000	5,250	
1.5 (150%)	70,000	3,500	
2.0 (200%)	52,500	2,625	
3.0 (300%)	35,000	1,750	Accessi

Min L for Corridor is the life pool at which the armour threshold falls exactly at the top of the 12–20×L corridor ($A = 20 \times L$). Y-stacking converts elemental DR from an armour-ceiling problem into a passive-investment problem: the armour remains within the corridor while additional passive nodes extend the elemental formula's reach. All thresholds scale proportionally lower at realistic hit sizes below 10,000.

At $Y = 1.0$ and $A = 80{,}000$ (representative of the upper corridor at $L = 5{,}000$, where $12\text{–}20 \times L = 60\text{k–}100\text{k}$), the outcome against the theoretical 10,000 ceiling hit is:

$$\begin{aligned}
\text{Physical DR:}\quad & \frac{80{,}000}{145{,}000} = 55.2\% \quad \Rightarrow \quad 6{,}500 \times 0.448 = 2{,}912 \text{ damage} \\
\text{Elemental DR:}\quad & \frac{80{,}000}{115{,}000} = 69.6\% \quad \Rightarrow \quad 3{,}500 \times 0.304 = 1{,}064 \text{ damage} \\
\text{Total damage taken:}\quad & 3{,}976 \quad (60.2\% \text{ total mitigation})
\end{aligned}$$

Pure-armour Smith at the same 80,000 armour takes $10{,}000 \times (1 - 80{,}000/180{,}000) =$ 5,556 damage against the same hit (44.4% mitigation). The armour-to-elemental configuration improves total mitigation by 15.8 percentage points at identical armour investment — achieved entirely through the elemental handling layer, with no additional armour cost. This gap is the structural benefit of the configuration: it extracts more mitigation from the same armour value by extending the formula's coverage to the converted elemental portion. At the theoretical ceiling of $A = 200{,}000$ the total mitigation rises to 78.9% (2,115 damage), approaching Shaman's 80.8%, but this armour value exceeds the achievable gear range and is noted for completeness only. The realistic build operates in the 40k–100k achievable armour band, where the benefit is a real 10–20 percentage point improvement over pure-armour Smith at the same investment level, not a path to matching conversion archetypes.

Veil of Night as the enabler. Once armour-to-elemental scaling is in place, elemental resistances become redundant: the armour formula handles the converted elemental component directly, and the 0% elemental resistance imposed by Veil of Night has no cost. The 50% increased maximum life is pure eHP gain — a 3,500 base life pool becomes 5,250. This raises the 12–20$\times L$ armour corridor from 42k–70k to 63k–105k, so the armour target scales upward naturally alongside the eHP gain. The two variables compound: higher eHP justifies higher armour, and higher armour extracts more value from the armour-to-elemental layer. Veil of Night's primary mechanism here is raising eHP first; the armour target follows through the corridor formula.

The correct build order is: acquire armour-to-elemental passive nodes first — without them, the elemental component is unmitigated and Veil of Night's 0% resistance is purely self-destructive. Once those nodes are in place, Veil of Night converts a resistance burden into a free eHP gain and natural corridor expansion.

The critical strike frontier. Smith of Kitava's structural advantage in critical strike map content adds a further dimension to this configuration. Blade Catcher converts critical strikes so they deal base damage (removing the critical multiplier) and defends those hits at 200% of armour rating — effectively $2A$ in the DR formula for crits. Combined with Internal Layer, the Smith is structurally unaffected by critical strikes in a way no other archetype in this paper achieves. In crit-heavy content, Smith's effective armour doubles on the hits that would otherwise be lethal, pushing physical crit DR well above what the base armour value alone suggests; the armour-to-elemental layer extends this advantage to the converted elemental components of those same crits.

The complete endgame Smith configuration — 12–20$\times L$ armour within the corridor, Y-stacking to reach 75%+ elemental DR, Veil of Night as a pure life multiplier, Blade Catcher providing crit immunity and $2A$ effective defence on crits — achieves a defensive profile that is inferior to Shaman in general endgame content but that dominates the critical strike map frontier. In that content specifically, the hits that would one-shot other archetypes land at base damage against double effective armour, and the elemental components of those crits are handled by the armour formula rather than by resistances. Smith occupies a narrower but genuine niche: the armour-based specialist for content where critical strikes are the primary threat.

11 Titan: The Armour Formula at Its Ceiling

Class Overview and Build Identity

The Titan is an ascendancy of the Warrior base class and the most direct expression of the "armour big and life big" paradigm in Path of Exile 2. Its ascendancy notables reinforce exactly what the armour formula rewards: higher effective armour values, bonuses to maximum life, and physical damage taken reduction from armour investment. The build's defensive theory is simple — invest in armour and life, and let the formula handle the rest.

The armour formula does not handle the rest. This is not a build-design failure unique to Titan; it is an expression of the formula's structural property. At the realistic achievable armour ceiling (40k–100k), DR against the endgame hit ceiling (H=10,000) falls between 28.6% and 50.0%. A Titan with 100,000 armour and 5,000 effective HP takes 5,000 damage from the endgame ceiling hit and survives by exactly 1 HP. There is no mechanism that suppresses the effective hit size, absorbs the post-DR remainder, or avoids the hit entirely. The character absorbs it all, reduced only by what the formula provides.

The Baseline Formula in Practice

Table 36: **Titan Survival at H=10,000 — Achievable Armour Range**

Armour	**Life Pool (L)**	**DR**	**Damage Taken**	**Survives?**
40,000	3,000	28.6%	7,143	No (−4,143)
60,000	4,000	37.5%	6,250	No (−2,250)
80,000	5,000	44.4%	5,556	No (−556)
100,000	5,001	50.0%	5,000	Yes (+1)

At the realistic upper armour corridor ($A = 80{,}000$ at $L = 5{,}000$, where $A = 16 \times L$), the Titan still cannot survive the endgame ceiling hit without 5,556 effective HP — 556 more than the pool allows. The 100,000 armour row requires 5,001 HP for a 1-HP survival margin. Compare to Shaman at 25,000 armour: 1,923 HP required. The Titan spends more passive points, more gear affixes, and more of its build budget to achieve worse survival at the hit ceiling.

Why Scaling Inputs Does Not Change the Output Structure

The Titan's implicit premise is that enough investment closes the gap between armour and conversion. The formula prevents this. Matching Shaman's 80.8% total mitigation through armour alone requires 421,000 armour rating (derived in the Character Sheet Dynamics section). The Titan's realistic ceiling is 100,000 armour, producing 50.0% DR. The 30.8 percentage-point gap is structural — it cannot be closed by adding passive points to armour clusters, because the formula's denominator grows faster than any practical armour budget can compensate.

The Colossal Capacity and Iron Skin ascendancy notables raise A and L simultaneously, which does compound well with the 12–20$\times L$ corridor: a larger L supports a larger armour target, and the armour notables help reach it. But raising both inputs does not alter the formula's structure. At A = 100,000 and L = 5,000: DR = 50.0%, VHP = 10,000, VHP/eHP = 2.0×. This is the armour formula's ceiling at realistic investment, regardless of how those values were reached.

Content Window and Realistic Use Case

The Titan's failure mode is specific to the top-tier endgame: hit sizes above H≈4,000, where DR falls below 50% and survival becomes a function of eHP more than armour investment. Below that threshold — which covers all of the levelling game, normal mapping, and early endgame encounters — 60,000–80,000 armour with a 4,000–5,000 HP pool provides functional and survivable DR. The Titan is a reliable platform through mid-game and early endgame. It is not a competitive ceiling for the hardest content.

This is the armour formula's argument made concrete: the formula is highly efficient against small-to-medium hits and degrades at exactly the hit sizes that define endgame survivability. The Titan, because it has no mechanism to change what the formula evaluates, reaches the formula's ceiling and stops. Every other archetype in this paper — including the Degenerate Gamblers — has found a way to exceed that ceiling.

Titan Positioning

Table 37: **Titan vs. Shaman: The Baseline Comparison**

Property	**Titan**	**Shaman**
Primary mechanism	Raw armour + life (formula inputs)	Conversion (suppresses
Armour at endgame ceiling	80,000–100,000	25,000
DR at H=10,000	44.4–50.0%	80.8%
Min HP for H=10k survival	5,001–5,556	1,923
VHP multiplier	2.0×	5.2×
Passive point cost (defensive)	High (armour + life clusters)	Minimal (ascendancy +
Unique item required	No	Yes (Cloak of Flame)
Chaos coverage	Armour DR only	Near-zero residual (res
Content ceiling	Mid-endgame	Full endgame
Verdict	**2.0× VHP floor — the reference point**	**5.2× VHP — 2.6×**

The Titan is the correct reference class for the armour formula's natural ceiling. Every other archetype in this paper achieves a VHP multiplier above 2.0×. The Titan's value is not that it underperforms — for most content it provides robust survivability — but that it performs exactly as the formula predicts, making it the cleanest possible baseline against which all other archetypes are measured.

12 Conclusion and Tier Summary

The hit-size spectrum analysis confirms and quantifies the paper's central insight: each archetype is solving the same underlying problem — inflating Virtual HP beyond what the armour formula alone provides — through a different structural mechanism. The tier ranking reflects how efficiently each mechanism converts passive investment into VHP/eHP multiplier.

1. **The armour formula has a structural floor.** A pure armour build at the optimal corridor ($A = 20 \times L$) achieves a 2.0× VHP multiplier. At the endgame hit ceiling (H=10,000), this requires 5,001 effective HP to survive. Every archetype above the Armour Pure baseline has found a way to exceed this floor.

2. **Conversion is the most passive-point-efficient mechanism.** Shaman achieves 5.2× VHP/eHP from a unique chest item and ascendancy notables, freeing the entire passive tree armour budget for damage and sustain. Infernalist Full reaches 6.8× by adding Chaos Inoculation and a further 5% conversion step — outperforming Shaman in raw mitigation while providing categorical chaos immunity.

3. **Absorption sidesteps the formula's hit-size degradation entirely.** Witchhunter's Sorcery Ward intercepts post-DR damage before it reaches life, making the survival ceiling independent of the formula's quadratic decay. Against a 10,000 raw hit at mid-endgame investment, the Witchhunter takes zero life damage. This is the paper's highest one-shot ceiling at comparable passive investment.
4. **The Degenerate Gamblers exceed the baseline even at worst case.** Invoker, Warbringer, and Tactician each operate through a probability chain (avoidance, block, or deflection). Their worst-case outcomes are not equivalent to Armour Pure — each carries armour-derived DR even when the primary avoidance layer fails. The trade-off is variance: expected performance is competitive with S-Tier; worst-case performance is above the baseline but below the deterministic archetypes.
5. **Smith of Kitava occupies a genuine niche.** The armour-to-elemental stacking and Blade Catcher mechanics improve on pure armour by 10–20 percentage points at realistic investment, and dominate the critical strike content frontier specifically. Smith is not a weaker Shaman; it is a different tool for a different content threat.
6. **Armour Pure / Titan is the baseline, not a viable ceiling.** At realistic gear levels, the formula produces 2.0× VHP. Every other archetype in this paper beats this figure at comparable or lower passive investment. The Titan is robust through mid-game content and should be evaluated on that axis, not on its endgame ceiling.

Tier Summary

The paper's central claim stands: at every tier above Armour Pure, the archetype is not a character with more armour — it is a character who has found a mechanism to change what the armour formula evaluates, or to make the formula irrelevant for a subset of hits. Path of Exile 2's defensive design space rewards this structural thinking and penalises the intuition that larger inputs to the same formula produce proportionally better outputs. They produce incrementally better outputs, subject to the formula's ceiling, and the ceiling is not high enough for the hardest content in the game.

Table 38: **Defensive Efficiency Tier Ranking — All Archetypes**

Tier	**Archetype**	**Mechanism**	**VHP/eHP**	**Min HP (H=10k)**
S	Witchhunter	SW absorption	∞ below ceiling	0 life dmg
S	Infernalist Full	Conversion + CI	6.8×	1,469
S	Shaman	Conversion	5.2×	1,923
A	Invoker	Avoidance + evasion-as-armour	>5× encounter	Conditional
A	Warbringer	Block + front-loaded mult.	6.6× exp / 2.0× wc	1,522 exp
A	Tactician	Deflect + capped DR	3.8× exp	2,625 exp
B	Smith of Kitava	Armour-to-ele + crit frontier	2.5×	3,976
C	Armour Pure / Titan	Raw armour + life	2.0×	5,001

VHP/eHP uses representative values at H=10,000. "Min HP" is the minimum effective HP to survive H=10,000. Warbringer and Tactician show expected (exp) values; their worst-case (wc) falls toward the armour-pure floor. Witchhunter's "0 life damage" reflects SW absorbing the full post-DR remainder of a 10,000 hit at mid-endgame investment; the survival ceiling is H≈14,485. Smith of Kitava achieves S-Tier performance in crit-map content specifically via Blade Catcher.